

CIRCLE

JEE EXERCISES

Exercise J-I

EQUATION OF CIRCLE

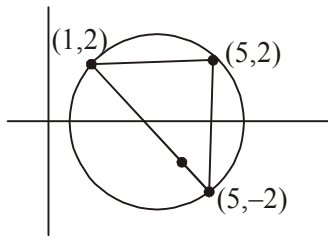
1. The radius of the circle passing through the points (1, 2), (5, 2) and (5, -2) is :

बिन्दुओं (1, 2), (5, 2) एवं (5, -2) से गुजरने वाले वृत्त की त्रिज्या है :

- (A) $5\sqrt{2}$ (B) $2\sqrt{5}$ (C) $3\sqrt{2}$ (D) $2\sqrt{2}$

Ans. D

Sol.



$$\text{diameter} = 4\sqrt{2} \Rightarrow r = 2\sqrt{2}$$

2. The lines $2x - 3y = 5$ and $3x - 4y = 7$ are diameters of a circle having area as 154 sq. units. Then the equation of the circle is : (Take $\pi = 22/7$)

रेखा $2x - 3y = 5$ तथा $3x - 4y = 7$ क्षेत्रफल 154 वर्ग इकाई वाले वृत्त के व्यास हैं। तब वृत्त का समीकरण है : ($\pi = 22/7$)

- (A) $x^2 + y^2 - 2x + 2y = 62$ (B) $x^2 + y^2 + 2x - 2y = 62$
(C) $x^2 + y^2 + 2x - 2y = 47$ (D) $x^2 + y^2 - 2x + 2y = 47$

Ans. D

Sol. Centre of circle = Point of intersection of diameters = (1, -1)

$$\text{Now area} = 154 \Rightarrow \pi r^2 = 154 \Rightarrow r = 7$$

Hence the equation of required circle is

$$(x - 1)^2 + (y + 1)^2 = 7^2 \Rightarrow x^2 + y^2 - 2x + 2y = 47$$

3. The equation of the circle passing through the points (1, 0) and (0, 1) and having the smallest radius is :

[AIEEE-2011]

उस वृत्त का समीकरण जो बिन्दुओं (1, 0) तथा (0, 1) से होकर जाता है तथा जिसकी त्रिज्या न्यूनतम है, हैं : [AIEEE 2011]

- (A) $x^2 + y^2 + x + y - 2 = 0$ (B) $x^2 + y^2 - 2x - 2y + 1 = 0$
(C) $x^2 + y^2 - x - y = 0$ (D) $x^2 + y^2 + 2x + 2y - 7 = 0$

Ans. C

Sol. (1, 0) and (0, 1) will be ends of diameter

So equation of circle

$$(x-1)(x-0) + (y-0)(y-1) \\ x^2 + y^2 - x - y = 0$$

INTERCEPT OF CIRCLE

4. The two lines through (2, 3) from which the circle $x^2 + y^2 = 25$ intercepts chords of length 8 units have equations:

- (A) $2x + 3y = 13$, $x + 5y = 17$ (B) $y = 3$, $12x + 5y = 39$
(C) $x = 2$, $9x - 11y = 51$ (D) none of these

बिन्दु (2, 3) से गुजरने वाली उन दो सरल रेखाओं के समीकरण जिन पर वृत्त $x^2 + y^2 = 25$, 8 इकाई लम्बाई के अन्तःखण्ड की जीवा

- (A) $2x + 3y = 13$, $x + 5y = 17$ (B) $y = 3$, $12x + 5y = 39$
(C) $x = 2$, $9x - 11y = 51$ (D) इनमें से कोई नहीं

Ans. B

Sol. Let slope of required line is m

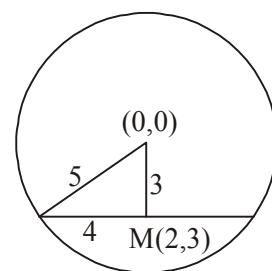
$$y - 3 = m(x - 2) \Rightarrow mx - y + (3 - 2m) = 0$$

length of $\perp$ from origin = 3

$$\Rightarrow 9 + 4m^2 - 12m = 9 + 9m^2 \Rightarrow 5m^2 + 12m = 0 \Rightarrow m = 0, -\frac{12}{5}$$

Hence lines are $y - 3 = 0 \Rightarrow y = 3$

$$y - 3 = -\frac{12}{5}(x - 2) \Rightarrow 5y - 15 = -12x + 24 \Rightarrow 12x + 5y = 39.$$



5. A variable circle passes through the fixed point A (p, q) and touches x-axis. The locus of the other end of the diameter through A is :

एक चर वृत्त स्थिर बिन्दु A (p, q) से गुजरता है। एवं x-अक्ष को स्पर्श करता है। A से गुजरने वाले व्यास के दूसरे सिरे का बिन्दुपथ है:

- (A) $(y-q)^2 = 4px$ (B) $(x-q)^2 = 4py$ (C) $(y-p)^2 = 4qx$ (D) $(x-p)^2 = 4qy$

Ans. D

Sol. Equation of circle having AB as diameter is

$$(x-p)(x-\alpha) + (y-q)(y-\beta) = 0$$

$$\text{or } x^2 + y^2 - (p+\alpha)x - (q+\beta)y + p\alpha + q\beta = 0 \quad \dots (i)$$

as it touches x-axis putting $y = 0$,

$$\text{we get } x^2 - (p+\alpha)x + p\alpha + q\beta = 0 \quad \dots (ii)$$

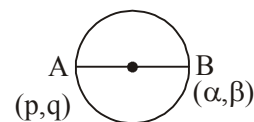
Since, circle (i) touches x-axis

Discriminant of equation (ii) = 0

$$\Rightarrow (p+\alpha)^2 = 4(p\alpha + q\beta) \Rightarrow (p-\alpha)^2 = 4q\beta$$

$\therefore$ Locus of B(α, β) is $(p-x)^2 = 4qy$

$$\text{or } (x-p)^2 = 4qy$$



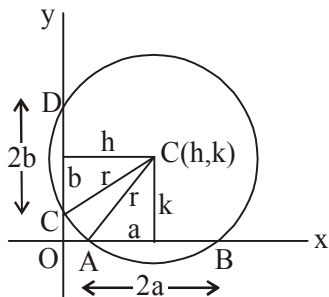
6. Two thin rods AB & CD of lengths $2a$ & $2b$ move along OX & OY respectively, when 'O' is the origin. The equation of the locus of the centre of the circle passing through the extremities of the two rods is :

दो पतली छड़े AB और CD जिनकी लम्बाई $2a$ और $2b$ है क्रमशः OX और OY अक्षों के अनुदिश गतिशील है। जहाँ 'O' मूलबिन्दु है, r के लocus के समीकरण ज्ञात करें।

- (A) $x^2 + y^2 = a^2 + b^2$ (B) $x^2 - y^2 = a^2 - b^2$ (C) $x^2 + y^2 = a^2 - b^2$ (D) $x^2 - y^2 = a^2 + b^2$

Ans. B

Sol. $h^2 + b^2 = r^2$ and $k^2 + a^2 = r^2 \Rightarrow h^2 - k^2 = a^2 - b^2$



$\therefore$ locus is $x^2 - y^2 = a^2 - b^2$

PARAMETRIC FORM OF CIRCLE

7. The centre of a set of circles, each of radius 3, lie on the circle $x^2 + y^2 = 25$. The locus of any point in the set is :

प्रत्येक 3 त्रिज्या के वृत्त के समुच्चय का केन्द्र वृत्त $x^2 + y^2 = 25$ पर स्थित है। समुच्चय किसी बिन्दु का बिन्दुपथ होगा :

- (A) $4 \leq x^2 + y^2 \leq 64$ (B) $x^2 + y^2 \leq 25$ (C) $x^2 + y^2 \geq 25$ (D) $3 \leq x^2 + y^2 \leq 9$

Ans. A

Sol. Let C lie on $x^2 + y^2 = 25$

Let S be the set of circles with radius 3 and center (h, k).

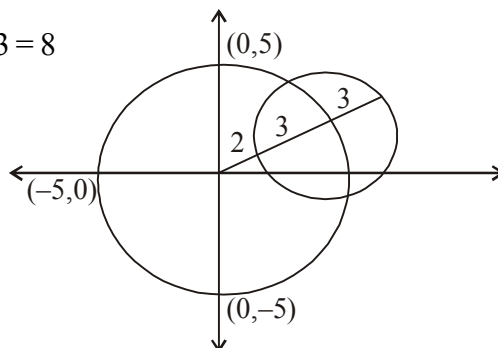
Any point P(h, k) on S will be 3 units from center C.

Distance from origin will be at least $5 - 3 = 2$. and at most $5 + 3 = 8$

$$\Rightarrow 2 \leq \sqrt{h^2 + k^2} \leq 8$$

$$4 \leq h^2 + k^2 \leq 64$$

Locus is $4 \leq x^2 + y^2 \leq 64$



8. If $\frac{x^2 + y^2}{x + y} = 4$, then all possible values of $(x - y)$ is given by :

यदि $\frac{x^2 + y^2}{x + y} = 4$ तो $(x - y)$ के सम्भावित मान है :

- (A) $[-2\sqrt{2}, 2\sqrt{2}]$ (B) $\{-4, 4\}$ (C) $[-4, 4]$ (D) $[-2, 2]$

Ans. C

Sol. We have $x^2 + y^2 = 4x + 4y$

$$\Rightarrow (x - 2)^2 + (y - 2)^2 = 8 = (2\sqrt{2})^2$$

Hence x, y lie on a circle with parametric coordinates

$$\begin{cases} x = 2 + 2\sqrt{2} \cos \theta \\ y = 2 + 2\sqrt{2} \sin \theta \end{cases} \Rightarrow x - y = 2\sqrt{2}(\cos \theta - \sin \theta)$$

$$\therefore x - y \in [-4, 4]$$

POSITION OF POINT W.R.T. A CIRCLE

9. The circle $x^2 + y^2 - 6x - 10y + c = 0$ does not intersect or touch either axis & the point $(1, 4)$ is inside the circle. Then the range of possible values of c is given by :

वृत्त $x^2 + y^2 - 6x - 10y + c = 0$ न तो अक्षों को काटता है और न ही स्पर्श करता है तथा बिन्दु $(1, 4)$ वृत्त के अन्दर स्थित हो, तो c के लिए c के

- (A) $c > 9$ (B) $c > 25$ (C) $c > 29$ (D) $25 < c < 29$

Ans. D

Sol. $S = x^2 + y^2 - 6x - 10y + k = 0$

Centre $C(3, 5)$, $r^2 = 9 + 25k = 34 - k$

Point $P(1, 4)$ lies inside the circle therefore

$$//S = -ve// -29 + k < 0 \text{ or } k < 29 \quad \dots(1)$$

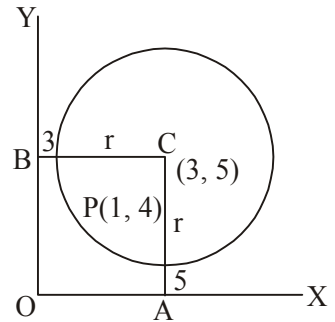
Since the circle neither touches nor intersects the axes of co-ordinates,

therefore from the figure it is clear that $r < 3$ (x coordinate of centre) and $r < 5$ (y co-ordinate of centre).

$$\text{Hence, } r < 3 \Rightarrow r^2 < 9$$

$$\text{or } 34 - k < 9 \text{ or } 25 < k \quad \dots(2)$$

from (1) and (2), we have $25 < k < 29$.



TANGENT AND NORMAL

10. The tangent lines to the circle $x^2 + y^2 - 6x + 4y = 12$ which are parallel to the line $4x + 3y + 5 = 0$ are given by :
 (A) $4x + 3y - 7 = 0, 4x + 3y + 15 = 0$ (B) $4x + 3y - 31 = 0, 4x + 3y + 19 = 0$
 (C) $4x + 3y - 17 = 0, 4x + 3y + 13 = 0$ (D) none of these

सरल रेखा $4x + 3y + 5 = 0$ के समान्तर वृत्त $x^2 + y^2 - 6x + 4y = 12$ की स्पर्श रेखाओं के समीकरण है :

- (A) $4x + 3y - 7 = 0, 4x + 3y + 15 = 0$ (B) $4x + 3y - 31 = 0, 4x + 3y + 19 = 0$
 (C) $4x + 3y - 17 = 0, 4x + 3y + 13 = 0$ (D) इनमें से कोई नहीं

Ans. B

Sol. Line parallel to given line $4x + 3y + 5 = 0$ is $4x + 3y + k = 0$
 This is tangent to $x^2 + y^2 - 6x + 4y - 12 = 0$

$$\left| \frac{12 - 6 + k}{5} \right| = 5 \Rightarrow 6 + k = \pm 25 \Rightarrow k = 19, -31$$

Hence required line $4x + 3y - 31 = 0, 4x + 3y + 19 = 0$

11. The condition so that the line $(x + g) \cos \theta + (y + f) \sin \theta = k$ is a tangent to $x^2 + y^2 + 2gx + 2fy + c = 0$ is :

सरल रेखा $(x + g) \cos \theta + (y + f) \sin \theta = k$ के वृत्त $x^2 + y^2 + 2gx + 2fy + c = 0$ की स्पर्श रेखा होने के लिए

- (A) $g^2 + f^2 = c + k^2$ (B) $g^2 + f^2 = c^2 + k$ (C) $g^2 + f^2 = c^2 + k^2$ (D) $g^2 + f^2 = c + k$

Ans. A

Sol.
$$p = \left| \frac{(-g + g) \cos \theta + (-f + f) \sin \theta - k}{\sqrt{\cos^2 \theta + \sin^2 \theta}} \right| = \sqrt{g^2 + f^2 - c} \Rightarrow g^2 + f^2 = c + k^2$$

12. The tangent to the circle $x^2 + y^2 = 5$ at the point $(1, -2)$ also touches the circle $x^2 + y^2 - 8x + 6y + 20 = 0$ at :

बिन्दु $(1, -2)$ पर वृत्त $x^2 + y^2 = 5$ की स्पर्श रेखा वृत्त $x^2 + y^2 - 8x + 6y + 20 = 0$ को किस बिन्दु पर स्पर्श करती है :

- (A) $(-2, 1)$ (B) $(-3, 0)$ (C) $(-1, -1)$ (D) $(3, -1)$

Ans. D

Sol. Equation of tangent $x - 2y = 5$

Let required point be (α, β)

$$\alpha x + \beta y - 4(x + \alpha) + 3(y + \beta) + 20 = 0 \Rightarrow x(\alpha - 4) + y(\beta + 3) - 4\alpha + 3\beta + 20 = 0$$

Comparing

$$\frac{\alpha - 4}{1} = \frac{\beta + 3}{-2} = \frac{4\alpha - 3\beta - 20}{5}$$

Similarly $(\alpha, \beta) \equiv (3, -1)$

POWER OF A POINT

13. The lengths of the tangents from any point of the circle $15x^2 + 15y^2 - 48x + 64y = 0$ to the two circles $5x^2 + 5y^2 - 24x + 32y + 75 = 0$, $5x^2 + 5y^2 - 48x + 64y + 300 = 0$ are in the ratio :

वृत्त $15x^2 + 15y^2 - 48x + 64y = 0$ पर स्थित किसी एक बिन्दु से वृत्त $5x^2 + 5y^2 - 24x + 32y + 75 = 0$ तथा $5x^2 + 5y^2 - 48x + 64y + 300 = 0$ पर डाली गई स्पर्श रेखाओं की लम्बाई का अनुपात है :

- (A) 1 : 2 (B) 2 : 3 (C) 3 : 4 (D) 1 : 4

Ans. A

Sol. Let any point $P(x_1, y_1)$ to the circle $x^2 + y^2 - \frac{16x}{5} + \frac{64y}{15} = 0 \Rightarrow x_1^2 + y_1^2 - \frac{16}{5}x_1 + \frac{64}{15}y_1 = 0$

Length of tangent from $P(x_1, y_1)$ to the circle are in ration

$$\frac{\sqrt{S_1}}{\sqrt{S_2}} = \frac{\sqrt{x_1^2 + y_1^2 - \frac{24}{5}x_1 + \frac{32}{5}y_1 + 15}}{\sqrt{x_1^2 + y_1^2 - \frac{48}{5}x_1 + \frac{64}{5}y_1 + 60}} = \frac{\sqrt{\frac{16}{5}x_1 - \frac{64}{15}y_1 - \frac{24}{5}x_1 + \frac{32}{5}y_1 + 15}}{\sqrt{\frac{16}{5}x_1 - \frac{64}{15}y_1 - \frac{48}{5}x_1 + \frac{64}{5}y_1 + 60}}$$

$$= \sqrt{\frac{-24x_1 + 32y_1 + 225}{-96x_1 + 128y_1 + 900}} = \sqrt{\frac{-24x_1 + 32y_1 + 225}{4(-24x_1 + 32y_1 + 225)}} = \frac{1}{2}$$

Alter :

Circle passes through origin (0, 0) so

$$\sqrt{\frac{S_1}{S_2}} = \sqrt{\frac{15}{60}} = \frac{1}{2}$$

14. If L_1 and L_2 are the length of the tangent from (0, 5) to the circles $x^2 + y^2 + 2x - 4 = 0$ and $x^2 + y^2 - y + 1 = 0$ then :

यदि L_1 तथा L_2 वृत्तों $x^2 + y^2 + 2x - 4 = 0$ तथा $x^2 + y^2 - y + 1 = 0$ पर बिन्दु (0, 5) से डाली गयी स्पर्श रेखाओं की लम्बाई है

- (A) $L_1 = 2L_2$ (B) $L_2 = 2L_1$ (C) $L_1 = L_2$ (D) $L_1^2 = L_2$

Ans. C

Sol. $L_1 = L_2 = \sqrt{21}$

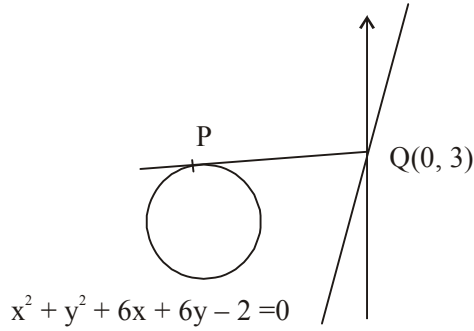
15. If the tangent at the point P on the circle $x^2 + y^2 + 6x + 6y = 2$ meets the straight line $5x - 2y + 6 = 0$ at a point Q on the y-axis, then the length of PQ is :

यदि वृत्त $x^2 + y^2 + 6x + 6y = 2$ पर स्थित बिन्दु P पर स्पर्श रेखा, रेखा $5x - 2y + 6 = 0$ को y अक्ष पर बिन्दु Q पर मिलती है, तो PQ की लम्बाई है :

- (A) 4 (B) $2\sqrt{5}$ (C) 5 (D) $3\sqrt{5}$

Ans. C

Sol.



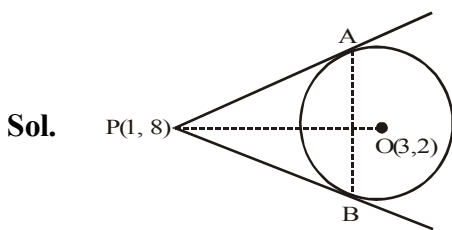
$$PQ = \sqrt{S_1} = 9 + 18 - 2$$

16. Tangents drawn from the point P(1, 8) to the circle $x^2 + y^2 - 6x - 4y - 11 = 0$ touch the circle at the points A and B. The equation of the circumcircle of the triangle PAB is : [JEE-2009]

बिन्दु P(1, 8) से वृत्त $x^2 + y^2 - 6x - 4y - 11 = 0$ पर खींची गई स्पर्श रेखायें वृत्त को A व B पर स्पर्श करती हैं। त्रिभुज PAB के

- (A) $x^2 + y^2 + 4x - 6y + 19 = 0$ (B) $x^2 + y^2 - 4x - 10y + 19 = 0$
(C) $x^2 + y^2 - 2x + 6y - 29 = 0$ (D) $x^2 + y^2 - 6x - 4y + 19 = 0$

Ans. B



The required circle is a circle described on OP as diameter.

DIRECTOR CIRCLE

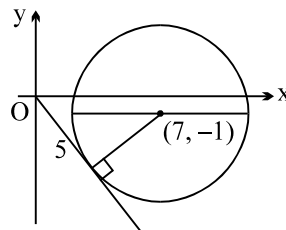
17. The angle between the two tangents from the origin to the circle $(x-7)^2 + (y+1)^2 = 25$ equals :

मूल बिन्दु से वृत्त $(x-7)^2 + (y+1)^2 = 25$ पर डाली गई स्पर्श रेखाओं के मध्य कोण होगा :

- (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{4}$

Ans. C

Sol. radius = 5
length of tangent = 5
∴ the figure is square
angle = 90°



CHORD WHOSE MID POINT IS GIVEN

18. The equation of the diameter of the circle $(x - 2)^2 + (y + 1)^2 = 16$ which bisects the chord cut off by the circle on the line $x - 2y - 3 = 0$ is :

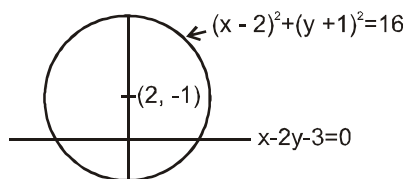
- (A) $x + 2y = 0$ (B) $2x + y - 3 = 0$ (C) $3x + 2y - 4 = 0$ (D) none

वृत्त $(x - 2)^2 + (y + 1)^2 = 16$ के व्यास का वह समीकरण जो वृत्त द्वारा सरल रेखा $x - 2y - 3 = 0$ पर काटी गई जीवा को लंबाई करता है :

- (A) $x + 2y = 0$ (B) $2x + y - 3 = 0$ (C) $3x + 2y - 4 = 0$ (D) इनमें से कोई नहीं

Ans. B

Sol. Required diameter is $\perp$ to given line.



$$\text{Hence } y + 1 = -2(x - 2) \Rightarrow 2x + y - 3 = 0.$$

19. The co-ordinates of the middle point of the chord cut off on $2x - 5y + 18 = 0$ by the circle $x^2 + y^2 - 6x + 2y - 54 = 0$ are :

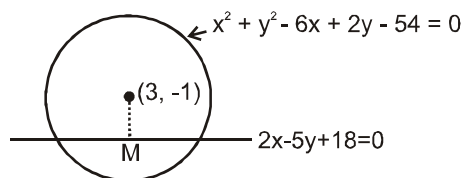
वृत्त $x^2 + y^2 - 6x + 2y - 54 = 0$ द्वारा सरल रेखा $2x - 5y + 18 = 0$ पर काटी गई जीवा के मध्य बिन्दु के निर्देशांक है :

- (A) (1, 4) (B) (2, 4) (C) (4, 1) (D) (1, 1)

Ans. A

Sol. Required point is foot of $\perp$

$$\frac{x - 3}{2} = \frac{y + 1}{-5} = -\left(\frac{6 + 5 + 18}{4 + 25}\right) = -1 \Rightarrow x = 1, y = 4$$



20. If the chord $y = mx + 1$ of the circle $x^2 + y^2 = 1$ subtends an angle of measure 45° at the major segment of the circle then value of m is :

- (A) $2 \pm \sqrt{2}$ (B) $-2 \pm \sqrt{2}$ (C) ± 1 (D) none of these

यदि वृत्त $x^2 + y^2 = 1$ की जीवा $y = mx + 1$ वृत्त के दीर्घ चाप पर 45° का कोण अन्तरित करता है, तब m का मान है :

- (A) $2 \pm \sqrt{2}$ (B) $-2 \pm \sqrt{2}$ (C) ± 1 (D) कोई नहीं

Ans. C

Sol. Correct option is C.

Given circle is $x^2 + y^2 = 1$

C(0,0) and radius = 1 and chord is $y = mx + 1$

$$\cos 45^\circ = \frac{CP}{CR}$$

CP = perpendicular distance from (0,0) to chord $y = mx + 1$

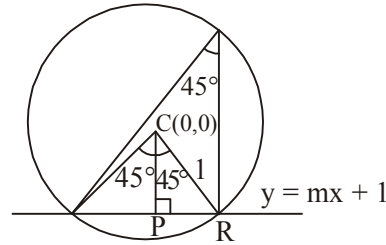
$$CP = \frac{1}{\sqrt{m^2 + 1}} \quad (CR = \text{radius} = 1)$$

$$\Rightarrow \cos 45^\circ = \frac{1}{\sqrt{m^2 + 1}}$$

$$\Rightarrow \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{m^2 + 1}}$$

$$\Rightarrow m^2 + 1 = 2$$

$$\Rightarrow m = \pm 1$$



21. Let C be the circle with centre (0, 0) and radius 3 units. The equation of the locus of the mid points of the chords of the circle C that subtend an angle of $\frac{2\pi}{3}$ at its centre is :

माना C केन्द्र (0, 0) एवं त्रिज्या 3 इकाई वाला वृत्त है। तब वृत्त C के केन्द्र पर $\frac{2\pi}{3}$ कोण अन्तरित करने वाली जीवा के मध्य बिन्दुओं

का लocus ज्ञात कीजिए।

- (A) $x^2 + y^2 = \frac{3}{2}$ (B) $x^2 + y^2 = 1$ (C) $x^2 + y^2 = \frac{27}{4}$ (D) $x^2 + y^2 = \frac{9}{4}$

Ans. D

Sol. Let AB be the chord subtending angle $2\pi/3$ at the centre C of circle

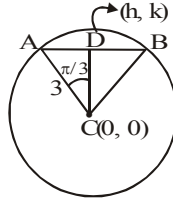
Now, $\angle ACD = \pi/3$

Let the coordinates of midpoint D be (h, k)

In $\triangle ACD$, $\cos \frac{\pi}{3} = \frac{CD}{CA}$

$\Rightarrow \frac{1}{2} = \frac{\sqrt{h^2 + k^2}}{3}$

$\Rightarrow x^2 + y^2 = \frac{9}{4}$, which is the required locus.



CHORD OF CONTACT

22. The chord of contact of the tangents drawn from a point on the circle, $x^2 + y^2 = a^2$ to the circle $x^2 + y^2 = b^2$ touches the circle $x^2 + y^2 = c^2$ then a, b, c are in :

वृत्त $x^2 + y^2 = a^2$ पर स्थित बिन्दु से वृत्त $x^2 + y^2 = b^2$ पर स्पर्श जीवा वृत्त $x^2 + y^2 = c^2$ को स्पर्श करती है जो a, b, c है :

- (A) A.P. (B) G.P. (C) H.P. (D) A.G.P.

Ans. B

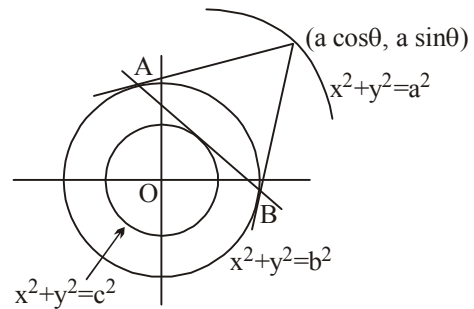
Sol. equation of chord of contact AB

$x a \cos \theta + y a \sin \theta = b^2$ (1)

this is tangent to the circle $x^2 + y^2 = c^2$

$\therefore$ perpendicular from $(0, 0)$ on (1) is equal to c

$\Rightarrow \frac{b^2}{\sqrt{a^2 \cos^2 \theta + b^2 \sin^2 \theta}} = c \Rightarrow b^2 = ac$



23. If tangent at $(1, 2)$ to the circle $c_1 : x^2 + y^2 = 5$ intersects the circle $c_2 : x^2 + y^2 = 9$ at A and B and tangents at A and B to the second circle meet at point C, then the co-ordinates of C are :

यदि वृत्त $c_1 : x^2 + y^2 = 5$ पर $(1, 2)$ पर स्पर्श रेखा वृत्त $c_2 : x^2 + y^2 = 9$ को A तथा B पर प्रदर्शित करती है एवं A तथा B पर खिंची गई स्पर्श रेखाओं का प्रतिच्छेद बिन्दु C पर मिलती है तो C के निर्देशांक है :

- (A) $(4, 5)$ (B) $(9/15, 18/5)$ (C) $(4, -5)$ (D) $(9/5, 18/5)$

Ans. D

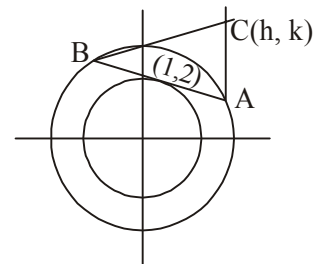
Sol. Tangent at $(1, 2)$ to the circle $x^2 + y^2 = 5$ is :

$x + 2y - 5 = 0$

chord of contact from $C(h, k)$ to $x^2 + y^2 = 9$ is :

$hx + ky - 9 = 0$

Comparing both equations, we get $\frac{h}{1} = \frac{k}{2} = \frac{9}{5}$



$$\Rightarrow (h, k) \equiv \left(\frac{9}{5}, \frac{18}{5} \right)$$

PAIR OF TANGENTS

24. Combined equation to the pair of tangents drawn from the origin to the circle $x^2 + y^2 + 4x + 6y + 9 = 0$ is :

वृत्त $x^2 + y^2 + 4x + 6y + 9 = 0$ पर मूल बिन्दु से डाली गई स्पर्श रेखा युग्म का संयुक्त समीकरण है :

- (A) $3(x^2 + y^2) = (x + 2y)^2$ (B) $2(x^2 + y^2) = (3x + y)^2$
 (C) $9(x^2 + y^2) = (2x + 3y)^2$ (D) $x^2 + y^2 = (2x + 3y)^2$

Ans. C

Sol. one of the tangent is y-axis i.e. $x = 0$ and only (C) satisfies $x = 0$

25. If $x = 3$ is the chord of contact of the circle $x^2 + y^2 = 81$, then the equation of the corresponding pair of tangents, is :

यदि $x = 3$ वृत्त $x^2 + y^2 = 81$ की स्पर्श जीवा है, तो इसके सापेक्ष स्पर्श रेखा युग्म का समीकरण है :

- (A) $x^2 - 8y^2 + 54x + 729 = 0$ (B) $x^2 - 8y^2 - 54x + 729 = 0$
 (C) $x^2 - 8y^2 - 54x - 729 = 0$ (D) $x^2 - 8y^2 = 729$

Ans. B

Sol. Equation of tangent to the circle at $A(3, 6\sqrt{2})$

$$3x + 6\sqrt{2}y = 81 \quad \dots(1)$$

and at B $(3, -6\sqrt{2})$

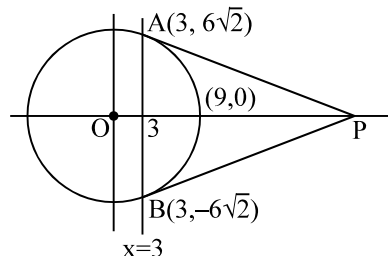
$$3x - 6\sqrt{2}y = 81 \quad \dots(2)$$

hence equation of the line pair

$$(x + 2\sqrt{2}y - 27)(x - 2\sqrt{2}y - 27) = 0$$

$$x^2 - 8y^2 - 27(x + 2\sqrt{2}y) - 27(x - 2\sqrt{2}y) + 729 = 0$$

$$x^2 - 8y^2 - 54x + 729 = 0$$



FAMILY OF CIRCLES

26. Intercept on the line $y = x$ by the circle $x^2 + y^2 - 2x = 0$, is AB. Equation of the circle on AB as a diameter is :

रेखा $y = x$ पर वृत्त $x^2 + y^2 - 2x = 0$ द्वारा बनाया गया अन्तःखण्ड AB है। AB को व्यास लेकर बनाये गये वृत्त का समीकरण है :

- (A) $x^2 + y^2 + x - y = 0$ (B) $x^2 + y^2 - x + y = 0$ (C) $x^2 + y^2 + x + y = 0$ (D) $x^2 + y^2 - x - y = 0$

Ans. D

Sol. Given, circle is $x^2 + y^2 - 2x = 0$ (i)

and line is $y = x$ (ii)

Putting $y = x$ in (i),

We get $2x^2 - 2x = 0 \Rightarrow x = 0, 1$

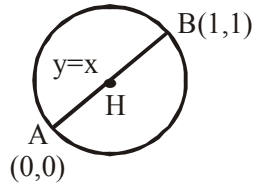
From (i), $y = 0, 1$

Let $A = (0, 0)$, $B = (1, 1)$

Equation of required circle is

$$(x - 0)(x - 1) + (y - 0)(y - 1) = 0$$

or $x^2 + y^2 - x - y = 0$



27. The length of the diameter of the circle which touches the x -axis at the point $(1, 0)$ and passes through the point $(2, 3)$ is : **[AIEEE-2012]**

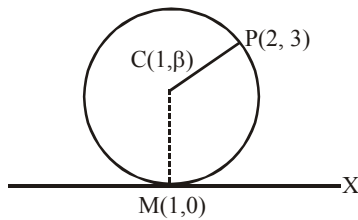
एक वृत्त जो x -अक्ष को बिन्दु $(1, 0)$ पर स्पर्श करता है तथा बिन्दु $(2, 3)$ से होकर जाता है तो व्यास की लम्बाई होगी :

[AIEEE-2012]

- (A) $\frac{5}{3}$ (B) $\frac{10}{3}$ (C) $\frac{3}{5}$ (D) $\frac{6}{5}$

Ans. B

Sol.



Let center of the circle be $C(1, \beta)$

$$\beta^2 = (2 - 1)^2 + (3 - \beta)^2$$

$$\Rightarrow \beta^2 = -6\beta + 10 + \beta^2$$

$$\Rightarrow \beta = \frac{5}{3}$$

$$\therefore r = \frac{5}{3}$$

$$\text{diameter} = \frac{10}{3}$$

28. The circle passing through the point $(-1, 0)$ and touching the y -axis at $(0, 2)$ also passes through the point : **[JEE-2011]**

बिन्दु $(-1, 0)$ से गुजरने वाले तथा y -अक्ष को $(0, 2)$ पर स्पर्श करने वाला वृत्त किस बिन्दु से गुजरता है :

[JEE-2011]

- (A) $\left(-\frac{3}{2}, 0\right)$ (B) $\left(-\frac{5}{2}, 0\right)$ (C) $\left(-\frac{3}{2}, \frac{5}{2}\right)$ (D) $(-4, 0)$

Ans. D

Sol. Family of circle which touches y-axis at $(0, 2)$ is

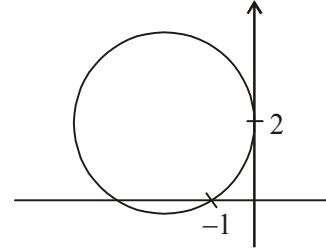
$$x^2 + (y - 2)^2 + \lambda x = 0$$

Passing through $(-1, 0)$

$$\Rightarrow 1 + 4 - \lambda = 0 \Rightarrow \lambda = 5$$

$$\therefore x^2 + y^2 + 5x - 4y + 4 = 0$$

which satisfy the point $(-4, 0)$.



29. A circle touches the x-axis and also touches the circle with centre at $(0, 3)$ and radius 2. The locus of the centre of the circle is :

- (A) an ellipse (B) a circle (C) a hyperbola (D) a parabola

एक वृत्त x-अक्ष को स्पर्श करता है तथा केन्द्र $(0, 3)$ व त्रिज्या 2 वाले वृत्त को भी स्पर्श करता है, तब वृत्त के केन्द्र का बिन्दुपथ है :

- (A) दीर्घ वृत्त (B) वृत्त (C) अतिपरवलय (D) परवलय

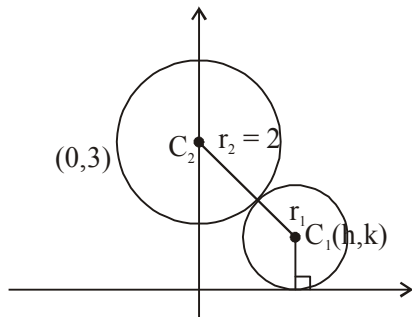
Ans. D

Sol. Let centre $\equiv (h, k)$; As $C_1 C_2 = r_1 + r_2$, (Given)

$$\Rightarrow \sqrt{(h-0)^2 + (k-3)^2} = |k + 2|$$

$$\Rightarrow h^2 = 5(2k - 1)$$

Hence locus, $x^2 = 5(2y - 1)$, which is parabola



POSITION OF TWO CIRCLES AND COMMON TANGENTS BETWEEN THEM

30. The centre of the circle passing through $(0, 0)$ and $(1, 0)$ and touching the circle $x^2 + y^2 = 9$ is :

$(0, 0)$ तथा $(1, 0)$ से गुजरने वाली एवं वृत्त $x^2 + y^2 = 9$ को स्पर्श करने वाले वृत्त का केन्द्र है :

- (A) $\left(\frac{1}{2}, \frac{1}{2}\right)$ (B) $\left(\frac{1}{2}, -\sqrt{2}\right)$ (C) $\left(\frac{3}{2}, \frac{1}{2}\right)$ (D) $\left(\frac{1}{2}, \frac{3}{2}\right)$

Ans. B

Sol. Correct option is B.

Let the equation of the circle be

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

which passes through (0,0) and (1,0).

$$\therefore c = 0 \text{ and } 1 + 2g + c = 0$$

$$\Rightarrow c = 0, g = -\frac{1}{2}$$

The circle $x^2 + y^2 + 2gx + 2fy + c = 0$ touches the circle $x^2 + y^2 = 9$

$$\therefore \sqrt{g^2 + f^2} = 3 \pm \sqrt{g^2 + f^2 - c} \quad [\because C_1 C_2 = I_1 \pm I_2]$$

$$\Rightarrow \sqrt{\frac{1}{4} + f^2} = 3 \pm \sqrt{\frac{1}{4} + f^2}$$

$$\Rightarrow 3 = 2\sqrt{\frac{1}{4} + f^2}$$

$$\Rightarrow f^2 = \frac{9}{4} - \frac{1}{4} = 2$$

$$\Rightarrow f = \pm\sqrt{2}$$

Hence, the coordinates of the centre are $\left(\frac{1}{2}, \pm\sqrt{2}\right)$

31. Let C be the circle with centre at (1, 1) and radius = 1. If T is the circle centred at (0, y), passing through origin and touching the circle C externally, then the radius of T is equal to : **[JEE(Main)2014]**

माना C एक वृत्त है जिसका केन्द्र (1, 1) पर है तथा त्रिज्या = 1 है। यदि T केन्द्र (0, y) वाला वृत्त है जो मूल बिन्दु से हो कर जाता है
T को C को बाह्य रूप से स्पर्श करता है, तो T की त्रिज्या बराबर है : **[JEE(Main)2014]**

- (A) $\frac{1}{2}$ (B) $\frac{1}{4}$ (C) $\frac{\sqrt{3}}{\sqrt{2}}$ (D) $\frac{\sqrt{3}}{2}$

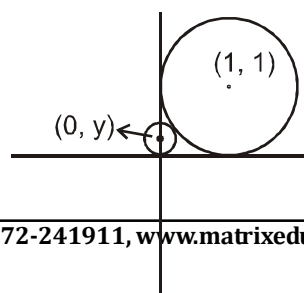
Ans. B

Sol. $c_1 (1, 1) \quad r_1 = 1$

$c_2 (0, y) \quad r_2 = |y|$

$$c_1 c_2 = r_1 + r_2$$

$$\sqrt{(1-0)^2 + (1-y)^2} = 1 + |y|$$



$$2 - 2y + y^2 = y^2 + 2|y| + 1$$

$$4|y| = 1$$

$$|y| = \frac{1}{4}$$

$$y = \frac{1}{4}$$

RADICAL AXIS AND ITS PROPERTIES

32. The equation of the common tangent to the circle $x^2 + y^2 - 4x - 6y - 12 = 0$ and $x^2 + y^2 + 6x + 18y + 26 = 0$ at their point of contact is :

वृत्त $x^2 + y^2 - 4x - 6y - 12 = 0$ और $x^2 + y^2 + 6x + 18y + 26 = 0$ के स्पर्श बिन्दु पर उभयनिष्ठ स्पर्श रेखा का समीकरण है :

(A) $12x + 5y + 19 = 0$ (B) $5x + 12y + 19 = 0$ (C) $5x - 12y + 19 = 0$ (D) $12x - 5y + 19 = 0$

Ans. B

Sol. Equation of common tangent at point of contact is $S_1 - S_2 = 0$

$$\Rightarrow 10x + 24y + 38 = 0 \Rightarrow 5x + 12y + 19 = 0$$

33. The locus of the centre of the circle which bisects the circumferences of the circles

$x^2 + y^2 = 4$ & $x^2 + y^2 - 2x + 6y + 1 = 0$ is :

(A) a straight line (B) a circle (C) a parabola (D) none of these

वृत्तों $x^2 + y^2 = 4$ और $x^2 + y^2 - 2x + 6y + 1 = 0$ की परिधियों को समद्विभाजित करने वाले वृत्त के केन्द्र का बिन्दु पथ है :

(A) सरल रेखा (B) वृत्त (C) परवलय (D) इनमें से कोई नहीं

Ans. A

Sol. Let required circle is $x^2 + y^2 + 2gx + 2fy + c = 0$

Hence common chord with $x^2 + y^2 - 4 = 0$

$$\text{is } 2gx + 2fy + c + y = 0$$

This is diameter of circle $x^2 + y^2 = 4$ hence $c = -4$.

Now again common chord with other circle

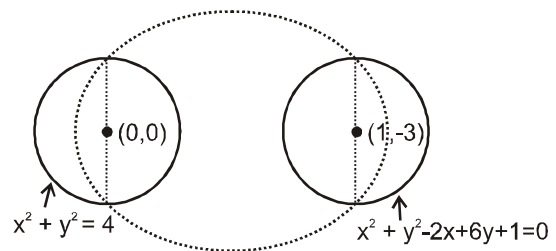
$$2x(g + 1) + 2y(f - 3) + (c - 1) = 0$$

This is diameter of $x^2 + y^2 - 2x + 6y + 1 = 0$

$$2(g + 1) - 6(f - 3) + 5 = 0$$

$$2g - 6f + 15 = 0$$

locus $2x - 3y - 15 = 0$ which is st. line.



34. The circumference of the circle $x^2 + y^2 - 2x + 8y - q = 0$ is bisected by the circle $x^2 + y^2 + 4x + 12y + p = 0$, then $p + q$ is equal to :

वृत्त $x^2 + y^2 - 2x + 8y - q = 0$ की परिधि को वृत्त $x^2 + y^2 + 4x + 12y + p = 0$ समद्विभाजित करता हो, तो $p + q$ का मान होगा:

- (A) 25 (B) 100 (C) 10 (D) 48

Ans. C

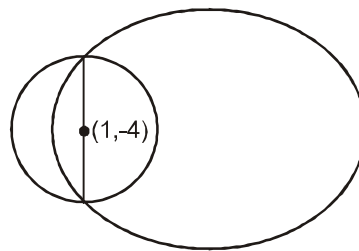
Sol. Common chord of given circle

$$6x + 4y + (p + q) = 0$$

This is diameter of $x^2 + y^2 - 2x + 8y - q = 0$

centre $(1, -4)$

$$6 - 16 + (p + q) = 0 \Rightarrow p + q = 10$$



35. If P and Q are the points of intersection of the circles $x^2 + y^2 + 3x + 7y + 2p - 5 = 0$ and $x^2 + y^2 + 2x + 2y - p^2 = 0$, then there is a circle passing through P, Q and $(1, 1)$ for : **[AIEEE-2009]**

- (A) all except one value of p (B) all except two values of p
(C) exactly one value of p (D) all values of p

यदि वृत्तों $x^2 + y^2 + 3x + 7y + 2p - 5 = 0$ तथा $x^2 + y^2 + 2x + 2y - p^2 = 0$ के प्रतिच्छेद बिन्दु P एवं Q हो, तो P, Q और $(1, 1)$ से गुजरने वाला एक वृत्त होगा यदि : **[AIEEE-2009]**

- (A) च के एक मान के अलावा सभी मानों के लिए (B) च के दो मानों के अलावा सभी मानों के लिए
(C) च के ठीक एक मान के लिए (D) च के सभी मानों के लिए

Ans. A

Sol. Eqⁿ of line PQ

$$x + 5y + 2p - 5 + p^2 = 0$$

P, Q and $(1, 1)$

will not lie on a circle of $(1, 1)$

Lies on the line

$$x + 5y + p^2 + 2p - 5 = 0$$

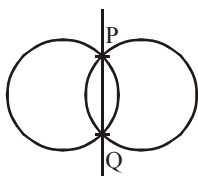
$$\Rightarrow 1 + 5 + p^2 + 2p - 5 = 0$$

$$p^2 + 2p + 1 = 0$$

$$\Rightarrow p = -1$$

Therefore there is a circle passing through P, Q and $(1, 1)$ for all values of p.

Except $p = -1$.



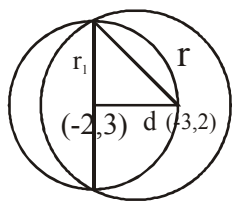
36. If one of the diameters of the circle, given by the equation, $x^2 + y^2 - 4x + 6y - 12 = 0$, is a chord of a circle S, whose centre is at $(-3, 2)$, then the radius of S is : **[JEE(Main)2016]**

यदि समीकरण $x^2 + y^2 - 4x + 6y - 12 = 0$ द्वारा प्रदत्त एक वृत्त का एक व्यास एक अन्य वृत्त S, जिसका केन्द्र $(-3, 2)$ है, की जीवा g की त्रिज्या है : **[JEE(Main)2016]**

- (A) 5 (B) 10 (C) $5\sqrt{2}$ (D) $5\sqrt{3}$

Ans. D

Sol.



$$r_1 = 5$$

$$d = \sqrt{50}$$

$$r = \sqrt{r_1^2 + d^2} = \sqrt{75} = 5\sqrt{3}$$

37. If the two circles, $x^2 + y^2 + 2g_1x + 2f_1y = 0$ and $x^2 + y^2 + 2g_2x + 2f_2y = 0$ touch each other then :

- (A) $r_1g_1 = f_2g_2$ (B) $\frac{f_1}{g_1} = \frac{f_2}{g_2}$ (C) $f_1f_2 = g_1g_2$ (D) none

यदि दो वृत्त $x^2 + y^2 + 2g_1x + 2f_1y = 0$ एवं $x^2 + y^2 + 2g_2x + 2f_2y = 0$ एक दूसरे को स्पर्श करते हैं, तो :

- (A) $r_1g_1 = f_2g_2$ (B) $\frac{f_1}{g_1} = \frac{f_2}{g_2}$ (C) $r_1f_2 = g_1g_2$ (D) इनमें से कोई नहीं

Ans. B

Sol. If two circles touch each other, then

$$C_1C_2 = r_1 + r_2$$

$$\sqrt{(-g_1 + g_2)^2 + (-f_1 + f_2)^2} = \sqrt{g_1^2 + f_1^2} + \sqrt{g_2^2 + f_2^2} \text{ squaring both sides}$$

$$-2g_1g_2 - 2f_1f_2 = 2\sqrt{(g_1^2 + f_1^2)(g_2^2 + f_2^2)} \Rightarrow (g_1f_2)^2 + (g_2f_1)^2 - 2g_1g_2f_1f_2 = 0 \Rightarrow \frac{g_1}{g_2} = \frac{f_1}{f_2}$$

ANGLE BETWEEN TWO CIRCLES & ORTHOGONALITY OF CIRCLES

38. Equation of the circle cutting orthogonally the three circles $x^2 + y^2 - 2x + 3y - 7 = 0$, $x^2 + y^2 + 5x - 5y + 9 = 0$ and $x^2 + y^2 + 7x - 9y + 29 = 0$ is :

- (A) $x^2 + y^2 - 16x - 18y - 4 = 0$ (B) $x^2 + y^2 - 7x + 11y + 6 = 0$
(C) $x^2 + y^2 + 2x - 8y + 9 = 0$ (D) none of these

तीन वृत्तों $x^2 + y^2 - 2x + 3y - 7 = 0$, $x^2 + y^2 + 5x - 5y + 9 = 0$ एवं $x^2 + y^2 + 7x - 9y + 29 = 0$ को लम्बकोणीय काटने वाले वृत्त का समीकरण है :

- (A) $x^2 + y^2 - 16x - 18y - 4 = 0$ (B) $x^2 + y^2 - 7x + 11y + 6 = 0$
(C) $x^2 + y^2 + 2x - 8y + 9 = 0$ (D) इनमें से कोई नहीं

Ans. A

Sol. $S_1 - S_2 = 0 \Rightarrow 7x - 8y + 16 = 0$
 $S_2 - S_3 = 0 \Rightarrow 2x - 4y + 20 = 0$
 $S_3 - S_1 = 0 \Rightarrow 9x - 12y + 36 = 0$

On solving centre (8, 9)

Length of tangent $= \sqrt{S_1} = \sqrt{64 + 81 - 16 + 27 - 7} = \sqrt{149} = (x - 8)^2 + (y - 9)^2 = 149$
 $= x^2 + y^2 - 16x - 18y - 4 = 0$

39. If a circle passes through the point (a, b) and cuts the circle $x^2 + y^2 = 4$ orthogonally, then the locus of its centre is:

यदि एक वृत्त बिन्दु (a, b) से गुजरता है तथा वृत्त $x^2 + y^2 = 4$ को लम्बकोणीय प्रतिच्छेद करता है, तब इसके केन्द्र का बिन्दुपथ है :

- (A) $2ax - 2by - (a^2 + b^2 + 4) = 0$ (B) $2ax + 2by - (a^2 + b^2 + 4) = 0$
(C) $2ax - 2by + (a^2 + b^2 + 4) = 0$ (D) $2ax + 2by + (a^2 + b^2 + 4) = 0$

Ans. B

Sol. Let the variable circle be

$$x^2 + y^2 + 2gx + 2fy + c = 0 \quad \dots (i)$$

Circle (i) cuts circle $x^2 + y^2 - 4 = 0$ orthogonally

$$\Rightarrow 2g \cdot 0 + 2f \cdot 0 = c - 4 \Rightarrow c = 4$$

Since circle (i) passes through (a, b)

$$\therefore a^2 + b^2 + 2ga + 2fb + 4 = 0$$

$\therefore$ Locus of centre $(-g, -f)$ is

$$2ax + 2by - (a^2 + b^2 + 4) = 0$$

Exercise J-II

SUBJECTIVE / INTEGER TYPE QUESTIONS

EQUATION OF CIRCLE

1. Determine the nature of the quadrilateral formed by four lines $3x + 4y - 5 = 0$; $4x - 3y - 5 = 0$; $3x + 4y + 5 = 0$ and $4x - 3y + 5 = 0$. Find the equation of the circle inscribed and circumscribing this quadrilateral.

रेखाओं $3x + 4y - 5 = 0$; $4x - 3y - 5 = 0$; $3x + 4y + 5 = 0$ तथा $4x - 3y + 5 = 0$ से बनने वाले चतुर्भुज की प्रकृति ज्ञात करें। इस चतुर्भुज के अन्तःवृत्त तथा परिवृत्त का समीकरण ज्ञात कीजिये।

Ans. square of side 2; $x^2 + y^2 = 1$; $x^2 + y^2 = 2$

Sol. $L_1 = 3x + 4y = 5 = m = -\frac{3}{4}$

$$L_2 = 4x - 3y = 5 = m = \frac{4}{3}$$

$$L_3 = 3x + 4y = -5 = m = -\frac{3}{4}$$

$$L_4 = 4x - 3y = -5 = m = \frac{4}{3}$$

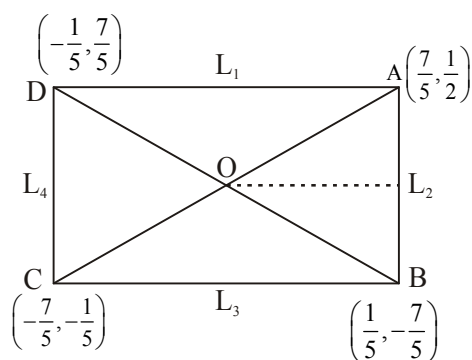
$$\text{Diagonal AC} = \frac{\frac{1}{5} + \frac{1}{5}}{\frac{7}{5} + \frac{7}{5}} = \frac{\frac{2}{5}}{\frac{14}{5}} = \frac{1}{7}$$

$$\text{Diagonal BD} = \frac{\frac{7}{5} + \frac{7}{5}}{-\frac{1}{5} - \frac{1}{5}} = \frac{\frac{14}{5}}{-\frac{2}{5}} = -7$$

$$M_{AC} \times M_{BD} = \frac{1}{7} \times -7$$

Diagonals are at right angle.

ABCD is square.



2. A circle $S = 0$ is drawn with its centre at $(-1, 1)$ so as to touch the circle $x^2 + y^2 - 4x + 6y - 3 = 0$ externally. Find the intercept made by the circle $S = 0$ on the coordinate axes.

वृत्त $x^2 + y^2 - 4x + 6y - 3 = 0$ के बाह्य स्पर्श करने वाले वृत्त $S = 0$ का समीकरण ज्ञात करें यदि वृत्त $S = 0$ का केन्द्र $(-1, 1)$ है।
वृत्त $S = 0$ द्वारा अक्षों पर काटे गये अन्तः खण्ड ज्ञात कीजियें।

Ans. zero, zero

Sol. Let the radius of the circle be r .

Since, they touch each other externally

Then, $C_1 C_2 = r_1 + r_2$

$$\Rightarrow \sqrt{(-1-2)^2 + (1+3)^2} = r+4$$

$$\Rightarrow r = 1$$

Therefore, equation of circle is $(x + 1)^2 + (y - 1)^2 = 1$

which touches both the co-ordinate axis.

Therefore, both the intercept are zero.

3 The line $lx + my + n = 0$ intersects the curve $ax^2 + 2hxy + by^2 = 1$ at the point P and Q. The circle on PQ as diameter passes through the origin. Prove that $n^2(a + b) = l^2 + m^2$.

वक्र $ax^2 + 2hxy + by^2 = 1$ को रेखा $lx + my + n = 0$ बिन्दु P तथा Q पर काटती है। PQ को व्यास लेकर खींचा गया वृत्त मूल बिन्दु $(0, 0)$ से गुजरता है यदि $n^2(a + b) = l^2 + m^2$.

Ans.

Sol. curve : $ax^2 + 2hxy + by^2 = 1$

$$\text{line : } \ln + my + n = 0 \Rightarrow \frac{\ln + my}{-n} = 1$$

homogenizing given curve with the given line

$$ax^2 + 2hxy + by^2 = \left(\frac{\ln + my}{-n} \right)^2$$

$$\left(a - \frac{l^2}{n^2}\right)x^2 + \left(2h - 2\frac{l}{n}\frac{m}{n}\right)xy + \left(b - \frac{m^2}{n^2}\right)y^2 = 0 \quad \text{_____}(1)$$

equation (1) give combine equation of $OA = OB$

as AB is diameter

$$\angle AOB = 90^\circ$$

Thus $m_1 m_2 = -1$

$m_1 \rightarrow$ slope of OB

$m_2 \rightarrow$ slope of OA

$$\left(\frac{y}{x}\right)^2 \left(b - \frac{m^2}{n^2}\right) + \frac{y}{x} \left(2h - \frac{2lm}{n}\right) + \left(a - \frac{l^2}{n^2}\right) = 0$$

Let $\frac{y}{x} = m$

This quad has 2 roots m_1 & m_2 .

we have $m_1 m_2 = -1$

$$\frac{a - \frac{l^2}{n^2}}{b - \frac{m^2}{n^2}} = -1$$

$$\Rightarrow (a + b) n^2 = l^2 + m^2$$

- 4 One of the diameters of the circle circumscribing the rectangle ABCD in order is $4y = x + 7$. If A & B are the points $(-3, 4)$ & $(5, 4)$ respectively, then find the area of the rectangle.

आयत ABCD के परिवृत का एक व्यास $4y = x + 7$ है। यदि A $(-3, 4)$ तथा B $(5, 4)$ हो तो आयत का क्षेत्रफल ज्ञात कीजिए।

Ans. 32 sq. unit

Sol. Clearly, the centre of the circle lies on the line $4y = x + 7$.

The circle passes through A $(-3, 4)$ and B $(5, 4)$

The slope of the segment joining A and B is zero.

$\therefore$ the slope of the perpendicular bisector of AB is not defined.

Hence the perpendicular bisector of AB will be parallel to the y-axis and will pass through

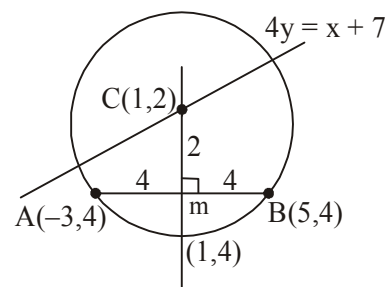
$$\left(\frac{-3+5}{2}, \frac{4+4}{2}\right) = (1, 4)$$

The equation of the perpendicular bisector is $x = 1$.

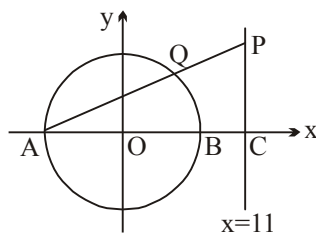
The intersection point of the perpendicular bisector and $4y = x + 7$ is $(1, 2)$.

$\therefore$ Centre = $(1, 2)$

Area of rectangle = $AB \times 2CM = 8 \times 2 \times 2 = 32$ square units.

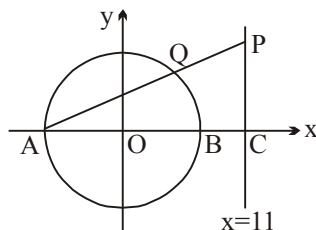


5. In the given figure, the circle $x^2 + y^2 = 25$ intersects the x-axis at the point A and B. The line $x = 11$ intersects the x-axis at the point C. Point P moves along the line $x = 11$ above the x-axis and AP intersects the circle at Q. Find



- The coordinates of the point P if the triangle AQB has the maximum area.
- The coordinates of the point P if Q is the middle point of AP.
- The coordinates of P if the area of the triangle AQB is $(1/4)^{\text{th}}$ of the area of the triangle APC.

दिये गये चित्र में वृत्त $x^2 + y^2 = 25$, x-अक्ष को बिन्दुओं A व B पर प्रतिच्छेद करता है। रेखा $x = 11$, x-अक्ष को बिन्दु C पर काटता है। बिन्दु P रेखा $x = 11$ पर x अक्ष के ऊपर विचरण करता है तथा AP वृत्त को Q बिन्दु पर काटता है। ज्ञात कीजिए।



- यदि त्रिभुज AQB का क्षेत्रफल अधिकतम हो तो बिन्दु P के निर्देशांक
- यदि AP का मध्य बिन्दु Q हो तो बिन्दु P के निर्देशांक
- यदि त्रिभुज AQB का क्षेत्रफल त्रिभुज APC के क्षेत्रफल का एक चौथाई हो तो P के निर्देशांक

Ans. (i) (11, 16), (ii) (11, 8), (iii) (11, 12)

Sol. (i) Area of $\triangle AQB|_{\max} = \frac{1}{2} \times AB \times QO = \frac{1}{2} \times 10 \times 5 = 25$

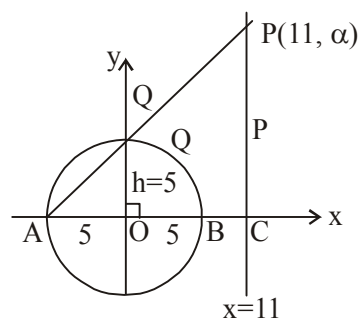
Equation of AQ $\Rightarrow y = x + 5$

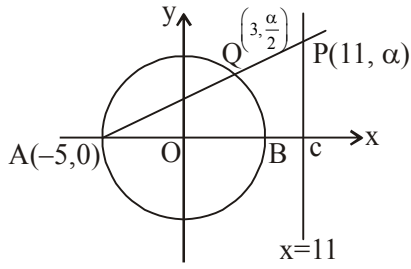
$\Rightarrow P(11, 16)$

(ii) Put $Q\left(3, \frac{\alpha}{2}\right)$ in $x^2 + y^2 = 25$

$\Rightarrow 9 + \frac{\alpha^2}{4} = 25 \Rightarrow \alpha^2 = 64 \Rightarrow \alpha = 8$

$\Rightarrow P(11, 8)$





$$(iii) \frac{\Delta APC}{\Delta AQB} = 4$$

$$\frac{AP^2}{AB^2} = 4$$

$$\frac{16^2 + \alpha^2}{10^2} = 4$$

$$\alpha^2 + 256 = 400$$

$$\alpha^2 = 144$$

$$\alpha = 12$$

INTERCEPT OF CIRCLE

- 6 Let L_1 be a straight line through the origin and L_2 be the straight line $x + y = 1$. If the intercepts made by the circle $x^2 + y^2 - x + 3y = 0$ on L_1 & L_2 are equal, then find the equation(s) which represent L_1 .

माना कि L_1 मूल बिन्दु से गुजरने वाली रेखा है तथा L_2 रेखा $x + y = 1$ है। यदि रेखाओं L_1 तथा $L_2 = 0$ पर वृत्त $x^2 + y^2 - x + 3y = 0$ द्वारा काटे गये अन्तःखण्ड बराबर हो तो L_1 को प्रदर्शित करने वाली रेखा के समीकरण को ज्ञात कीजिए।

Ans. $x - y = 0$; $x + 7y = 0$

Sol. $\therefore L_1$ passes through origin.

Let equation of L_1 $y = mx$

Intercepts made by L_1 and L_2 are equal

$\Rightarrow L_1$ and L_2 are at the same distance from centre of circle $x^2 + y^2 - x + 3y = 0$

centre of circle $x^2 + y^2 - x + 3y = 0$

$$x^2 + y^2 - x + 3y = 0$$

$$x^2 - x + \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 + y^2 + 3y + \left(\frac{3}{2}\right)^2 - \left(\frac{3}{2}\right)^2 = 0$$

$$\left(x - \frac{1}{2}\right)^2 + \left(y + \frac{3}{2}\right)^2 = \left(\frac{\sqrt{10}}{2}\right)^2$$

$$\therefore \text{Centre of circle } \left(\frac{1}{2}, -\frac{3}{2}\right)$$

Distance of a point (x_1, y_1) from line $Ax + By + C = 0$ is given by

$$d = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$$

$$\therefore \text{Distance of } \left(\frac{1}{2}, -\frac{3}{2}\right) \text{ from line } L_1 \text{ and } L_2 \text{ are}$$

$$d_1 = \frac{\left|m\left(\frac{1}{2}\right) + \frac{3}{2}\right|}{\sqrt{m^2 + 1}}$$

$$d_2 = \frac{\left|\frac{1}{2} + \left(\frac{3}{2}\right) - 1\right|}{\sqrt{1 + 1}}$$

$$d_1 = d_2$$

$$\Rightarrow \frac{m-3}{2\sqrt{m^2+1}} = \frac{3}{\sqrt{2}}$$

Squaring both sides we get

$$\frac{(m-3)^2}{4(m^2+1)} = \frac{9}{2}$$

$$(m-3)^2 = 8(m^2+1)$$

$$m^2 + 9 + 6m = 8m^2 + 8$$

$$7m^2 - 6m - 1 = 0$$

$$m = \frac{6 \pm \sqrt{36 + 28}}{14} = \frac{6 \pm 8}{14} = +1, -\frac{1}{7}$$

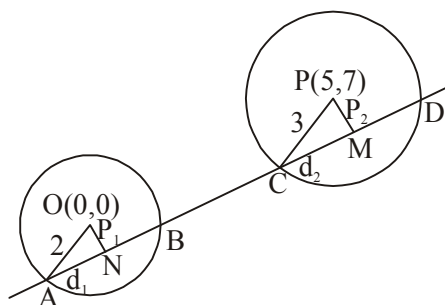
$$\therefore L_1 : -y + x = 0, x + 7y = 0.$$

- 7 Find the equation of a line with gradient 1 such that the two circles $x^2 + y^2 = 4$ and $x^2 + y^2 - 10x - 14y + 65 = 0$ intercept equal length on it.

उस रेखा का समीकरण ज्ञात करें जिसकी प्रवणता 1 हो तथा दो वृत्त $x^2 + y^2 = 4$ तथा $x^2 + y^2 - 10x - 14y + 65 = 0$ उस रेखा पर समान लम्बाई के अन्तः खण्ड काटते हों।

Ans. $2x - 2y - 3 = 0$

Sol. Equation of the circles $x^2 + y^2 = 4$ and $x^2 + y^2 - 10x - 14y + 65 = 0$ have centres, radius as $(0,0)$, 2 and $(5,7)$, 3 respectively.



Let equation of line be $y - x = c$... (i)

Perpendicular from $(0,0)$ on line (i) is $\left| \frac{-c}{\sqrt{2}} \right| = \frac{c}{\sqrt{2}}$

In $\triangle AON$, $\sqrt{2^2 - \left(\frac{c}{\sqrt{2}} \right)^2} = AN$

Similarly in $\triangle CPM$, $\sqrt{3^2 - \left(\frac{2-c}{\sqrt{2}} \right)^2} = CM$

Given, $AB = CD$

$\Rightarrow AN = CM$

$$\Rightarrow 4 - \frac{c^2}{2} = 9 - \frac{(2-c)^2}{2}$$

$$\Rightarrow c = -\frac{3}{2}$$

Therefore, equation of line :

$$y = x - \frac{3}{2}$$

$$\Rightarrow 2x - 2y - 3 = 0$$

- 8 Consider a circle S with centre at the origin and radius 4. Four circles A, B, C and D each with radius unity and centres $(-3, 0)$, $(-1, 0)$, $(1, 0)$ and $(3, 0)$ respectively are drawn. A chord PQ of the circle S touches the circle B and passes through the centre of the circle C. If the length of this chord can be expressed as $\sqrt{x}$, find x.

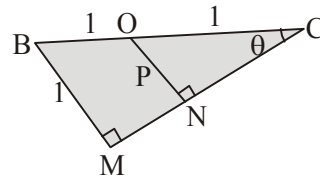
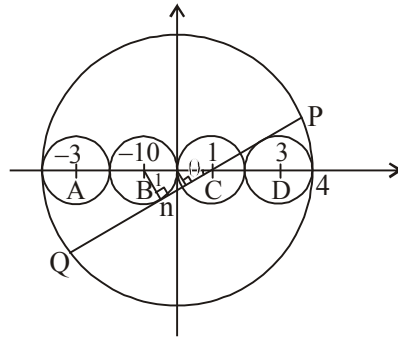
एक वृत्त S का केन्द्र मूल बिन्दु तथा त्रिज्या 4 है। चार वृत्त A, B, C तथा D जिनकी त्रिज्या 1 व केन्द्र क्रमशः $(-3, 0)$, $(-1, 0)$, $(1, 0)$ तथा $(3, 0)$ है खींचे गये हैं। वृत्त S की एक जीवा PQ वृत्त B को स्पर्श करती है तथा वृत्त C के केन्द्र से गुजरती है। यदि इस जीवा की लम्बाई $\sqrt{x}$ है तो x ज्ञात कीजिए।

Ans. 63

Sol. $\sin \theta = \frac{1}{2} = \frac{ON}{1} \Rightarrow ON = \frac{1}{2} = P$

$$\text{Length of chord} = 2\sqrt{r^2 - p^2}$$

$$PQ = 2\sqrt{16 - \frac{1}{4}} = \sqrt{63}$$



PARAMETRIC FORM OF CIRCLE

9. Real number x, y satisfies $x^2 + y^2 = 1$. If the maximum and minimum value of the expression $z = \frac{4-y}{7-x}$ are M and m respectively, then find the value $(2M + 6m)$.

वास्तविक संख्यायें x व y , $x^2 + y^2 = 1$ को संतुष्ट करती है। यदि $z = \frac{4-y}{7-x}$ का अधिकतम व न्यूनतम मान क्रमशः M व m है तो

$(2M + 6m)$ का मान ज्ञात कीजिए।

Ans. 4

Sol. $x^2 + y^2 = 1$

$$Z = \frac{y-4}{x-7}$$

In this slope from the point $(7,4)$:

$$\tan \theta_2 = \frac{4}{7}$$

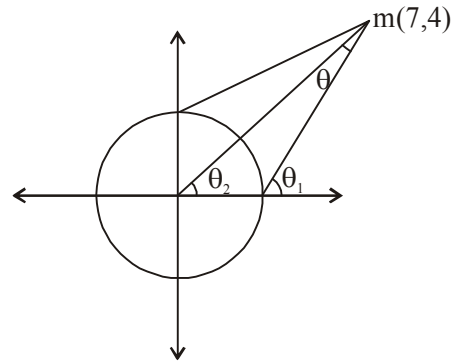
$$\tan \theta = \tan(\theta_1 - \theta_2) = \frac{m - \tan \theta_2}{1 + m \tan \theta_2}$$

$$\left| \frac{r}{\ell_{\text{tangent}}} \right| = \frac{m - \frac{4}{7}}{1 + m \times \frac{4}{7}}$$

$$\Rightarrow \pm \frac{1}{8} = \frac{m - \frac{4}{7}}{1 + \frac{4m}{7}} = \frac{7m - 4}{7 + 4m}$$

$$\therefore M = \frac{3}{4} \text{ and } m = \frac{5}{12}$$

$$\therefore 2M + 6m = \frac{2 \times 3}{4} + \frac{6 \times 5}{12} = 4$$



10 Given that $x^2 + y^2 = 14x + 6y + 6$, find the largest possible value of the expression $E = 3x + 4y$.

दिया गया है कि $x^2 + y^2 = 14x + 6y + 6$, तो व्यंजक $E = 3x + 4y$ का अधिकतम सम्भावित मान ज्ञात कीजिए।

Ans. 73

Sol. $(x-7)^2 + (y-3)^2 = 64 = 8^2$, a circle with centre $(7, 3)$ and radius 8

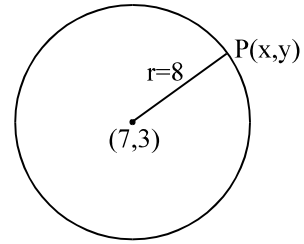
now $\frac{x-7}{\cos \theta} = \frac{y-3}{\sin \theta} = 8$

$\left. \begin{array}{l} x = 8 \cos \theta + 7 \\ y = 8 \sin \theta + 3 \end{array} \right\}$ lies on the circle for some θ

$\therefore E = 3x + 4y = 3(8 \cos \theta + 7) + 4(8 \sin \theta + 3)$

$E = (21 + 12) + 24 \cos \theta + 32 \sin \theta$
 $= 33 + 8(3 \cos \theta + 4 \sin \theta)$

$\therefore E_{\max} = 33 + 40 = 73 \text{ Ans.}$



POSITION OF POINT WRT A CIRCLE

11. Find the shortest distance from the point $M(-7, 2)$ to the circle $x^2 + y^2 - 10x - 14y - 151 = 0$.

बिन्दु $M(-7, 2)$ से वृत्त $x^2 + y^2 - 10x - 14y - 151 = 0$ की न्यूनतम दूरी ज्ञात कीजिए।

Ans. 2

Sol. $S = x^2 + y^2 - 10x - 14y - 151 = 0$

Now putting the values of points $(-7, 2)$ in the equation to check whether the given point is on the circle, inside the circle or outside the circle.

$S = -7^2 + 2^2 - 10 \times (-7) - 14 \times (2) - 151 = 0$

$S = 49 + 4 + 70 - 28 - 151 = -56$

$S = -56 < 0$

Therefore, $M(-7, 2)$ does not lie on the circle.

The general equation of the circle is,

$ax^2 + by^2 + 2gx + 2fy + c = 0$

Now, $g = 5, f = 7, c = -151$

The radius of the circle is,

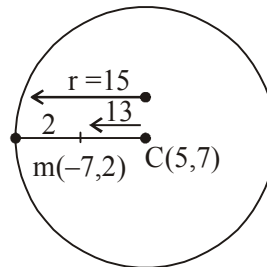
$r = \sqrt{g^2 + f^2 - c}$

$= \sqrt{(5^2 + (7)^2 - (-151))}$

$\sqrt{25 + 49 + 151} = \sqrt{225} = 15$

The centre of the circle is, $C = (-g, -f) = (5, 7)$

Now, $CM = \sqrt{(5+7)^2 + (7-2)^2}$



$$\Rightarrow CM = \sqrt{12^2 + 5^2}$$

$$\Rightarrow CM = \sqrt{144 + 25}$$

$$\Rightarrow CM = \sqrt{169} = 13$$

The shortest distance is, $r - CM = 15 - 13 = 2$

Hence the shortest distance is 2 units.

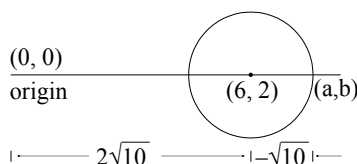
12. Find the co-ordinate of the point on the circle $x^2 + y^2 - 12x - 4y + 30 = 0$, which is farthest from the origin.

वृत्त $x^2 + y^2 - 12x - 4y + 30 = 0$ पर मूल से अधिकतम दूरी का बिन्दु ज्ञात कीजिए।

Ans. (9, 3)

Sol. radius $= 36 + 4 - 30 = \sqrt{10}$

Use section formula



13. The straight line $2x - 3y = 1$ divides the circular region $x^2 + y^2 \leq 6$ into two parts. If

$S = \left[\left(2, \frac{3}{4} \right), \left(\frac{5}{2}, \frac{3}{4} \right), \left(\frac{1}{4}, -\frac{1}{4} \right), \left(\frac{1}{8}, \frac{1}{4} \right) \right]$, then the number of point(s) in S lying inside the smaller part is :

[JEE-2011]

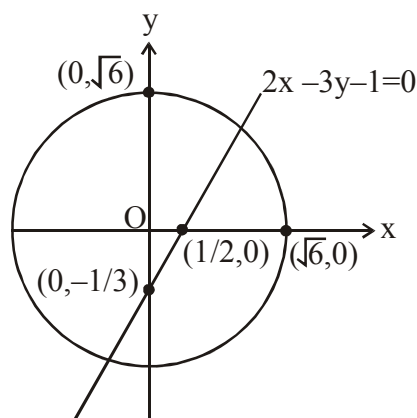
रेखा $2x - 3y = 1$ वृत्तीय क्षेत्र $x^2 + y^2 \leq 6$ को दो भागों में विभाजित करती है यदि $S = \left[\left(2, \frac{3}{4} \right), \left(\frac{5}{2}, \frac{3}{4} \right), \left(\frac{1}{4}, -\frac{1}{4} \right), \left(\frac{1}{8}, \frac{1}{4} \right) \right]$,

तो समुच्चय S में स्थित बिन्दुओं की संख्या जो छोटे भाग में स्थित है।

[JEE-2011]

Ans. 2

Sol.



If the point lies inside the smaller part, then origin and point should give opposite signs w.r.t. line & point should lie inside the circle.

for origin : $2 \times 0 - 3 \times 0 - 1 = -1$ (-ve)

for $(2, \frac{3}{4})$: $2 \times 2 - 3 \times \frac{3}{4} - 1$

$= \frac{3}{4}$ (+ve); point lies inside the circle

for $(\frac{5}{2}, \frac{3}{4})$: $2 \times \frac{5}{2} - 3 \times \frac{3}{4} - 1 = \frac{7}{4}$ (+ve); point lies outside the circle

For $(\frac{1}{4}, -\frac{1}{4})$: $2 \times \frac{1}{4} - 3 \times (-\frac{1}{4}) - 1 = \frac{1}{4}$ (+ve); point lies inside the circle

For $(\frac{1}{8}, \frac{1}{4})$: $2 \times \frac{1}{8} - 3 \times (\frac{1}{4}) - 1 = -\frac{3}{2}$ (-ve); point lies inside the circle.

$\therefore$ 2 points lie inside smaller part.

TANGENT AND NORMAL

- 14 A circle passes through the points $(-1, 1)$, $(0, 6)$ and $(5, 5)$. Find the points on the circle the tangents at which are parallel to the straight line joining origin to the centre.

एक वृत्त बिन्दुओं $(-1, 1)$, $(0, 6)$ तथा $(5, 5)$ से गुजरता है। वृत्त पर स्थित उन बिन्दुओं को ज्ञात करें जिन पर खींची गई स्पर्श रेखायें मूल बिन्दु को केन्द्र से मिलाने वाली रेखा के समान्तर होंगे।

Ans. $(5, 1)$ & $(-1, 5)$

Sol. Let the equation of circle be $x^2 + y^2 + 2gx + 2fy + c = 0$

It passes through $(-1, 1)$

$$\Rightarrow 1 + 1 - 2g + 2f + c = 0$$

$$\Rightarrow -2g + 2f + c = -2 \quad \text{_____ (i)}$$

It passes through $(0, 6)$

$$\Rightarrow 0 + 36 + 0 + 12f + c = 0$$

$$\Rightarrow 12f + c = -36 \quad \text{_____ (ii)}$$

It passes through $(5, 5)$

$$\Rightarrow 25 + 25 + 10g + 10f + c = 0$$

$$\Rightarrow 10g + 10f + c = -50 \quad \text{_____ (iii)}$$

Subtracting (iii) from (i) we get

$$-2g - 10f = 34 \quad \text{_____ (iv)}$$

subtracting (iii) from (ii) we get

$$-10g + 2f = 14 \quad \text{_____ (v)}$$

Solving (iv) and (v)

$$\Rightarrow g = -2, f = -3$$

Substituting g and f in (i) we get $c = 0$

So the equation of circle is

$$x^2 + y^2 - 4x - 6y = 0 \quad \text{_____ (vi)}$$

Centre of the circle is (2,3)

$$\text{Slope of line joining centre to origin} = \frac{3-0}{2-0} = \frac{3}{2}$$

Equation of line passing through the centre and perpendicular to line joining centre to origin is

$$y - 3 = \frac{-2}{3}(x - 2)$$

$$3y - 9 = -2x + 4$$

$$2x + 3y = 13 \quad \text{_____ (vii)}$$

Tangents will be parallel to the line joining the origin to the centre at the points where (vii) cuts the circle.

Substituting y from (vii) in (vi) we get

$$13x^2 - 52x - 65 = 0$$

$$x^2 - 4x - 5 = 0$$

$$x^2 - 5x + x - 5 = 0$$

$$x(x - 5) + (x - 5) = 0$$

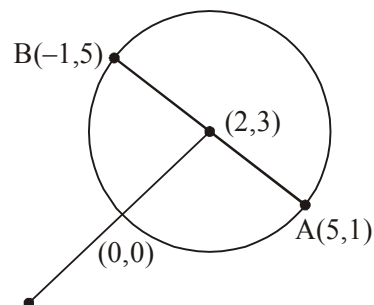
$$(x + 1)(x - 5) = 0$$

$$x = -1, 5$$

Substituting x in (vii) we get

$$y = 5, 1$$

So the point of contact of tangents are (-1,5) and (5,1)



- 15 A circle is drawn with its centre on the line $x + y = 2$ to touch the line $4x - 3y + 4 = 0$ and pass through the point (0, 1). Find its equation.

रेखा $x + y = 2$ पर केन्द्र लेते हुये वृत्त इस प्रकार से खींचा गया है कि वह $4x - 3y + 4 = 0$ को स्पर्श करता है तथा बिन्दु $(0, 1)$ से गुजरता है। वृत्त का समीकरण ज्ञात करें।

Ans. $x^2 + y^2 - 2x - 2y + 1 = 0$ OR $x^2 + y^2 - 42x + 38y - 39 = 0$

Sol. Let the circle be

$$(x - h)^2 + (y - k)^2 = r^2 \quad \dots(1)$$

We have $h + k = 2 \quad \dots(2)$

$$(0 - h)^2 + (1 - k)^2 = r^2 \quad \dots(3)$$

$$\text{and } (4h - 3k + 4)^2 = 25r^2 \quad \dots(4)$$

from (3) and (4)

$$(4h - 3k + 4)^2 = 25[h^2 + (1 - k)^2]$$

$$\Rightarrow 9h^2 + 16k^2 + 24hk - 32h - 26k + 9 = 0 \quad \dots(5)$$

Using (2) and (5) reduce to

$$9h^2 + 16(2 - h)^2 + 24h(2 - h) - 32h - 25(2 - h) + 9 = 0$$

$$\Rightarrow h^2 - 22h + 21 = 0 \Rightarrow h = 1, 21 \Rightarrow k = 1, -19$$

$$\Rightarrow r^2 = h^2 + (1 - k)^2 = 1 \Rightarrow 21^2 + 20^2 = 1, 841$$

Two circle are

$$(x - 1)^2 + (y - 1)^2 = 1 \text{ and } (x - 21)^2 + (y + 19)^2 = 841$$

- 16 Find the locus of the middle points of portions of the tangents to the circle $x^2 + y^2 = a^2$ terminated by the coordinate axes.

वृत्त $x^2 + y^2 = a^2$ की उन स्पर्श रेखाओं के मध्य बिन्दुओं का बिन्दुपथ ज्ञात करें जिनके सिरे अक्षों पर स्थित हैं।

Ans. $a^2(x^2 + y^2) = 4x^2y^2$

Sol. equation of tangent of $x^2 + y^2 = a^2$ at (x_1, y_1)

$$xx_1 + yy_1 = a^2$$

$$x_1 = a \cos \theta$$

$$y_1 = a \sin \theta$$

$$x a \cos \theta + y a \sin \theta = a^2$$

$$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{a} = 1$$

$$\text{intercept on x-axis } \frac{a}{\cos \theta} = 1$$

intercept on y-axis $\frac{a}{\sin \theta} = m$

Let mid-point of AB be (h, k)

$$h = \frac{1+0}{2}, 1 = 2h$$

$$k = \frac{m+0}{2}, m = 2k$$

$$\frac{a}{\cos \theta} = 2h$$

$$\frac{a}{\sin \theta} = 2k$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\left(\frac{a}{2h}\right)^2 + \left(\frac{a}{2k}\right)^2 = 1$$

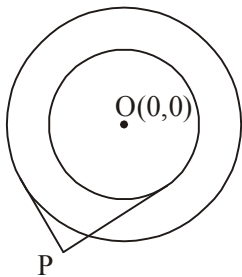
$$\text{So locus } \frac{a^2}{x^2} + \frac{a^2}{y^2} = 4$$

- 17 Tangents are drawn to the concentric circles $x^2 + y^2 = a^2$ and $x^2 + y^2 = b^2$ at right angle to one another. Show that the locus of their point of intersection is a 3rd concentric circle. Find its radius.

वृत्त $x^2 + y^2 = a^2$ तथा $x^2 + y^2 = b^2$ पर परस्पर लम्बवत् स्पर्श रेखायें खींची गई हैं। प्रदर्शित करो कि उनके प्रतिच्छेद बिन्दु का बिन्दुपथ तृतीय सहकेन्द्रीय वृत्त होगा। उस वृत्त की त्रिज्या भी ज्ञात कीजिए।

Ans. $x^2 + y^2 = a^2 + b^2$; $r = \sqrt{a^2 + b^2}$

Sol. Both circles are concentric at the origin (see fig.). From the note given under



$$y = mx + a\sqrt{1+m^2}$$

$$y = \frac{-1}{m}x + b\sqrt{a + \frac{1}{m^2}}$$

are tangents to the given circles and are at right angles. Therefore,

$$(y - mx^2) + (x + my)^2 = (a^2 + b^2)(1 + m^2)$$

$$\Rightarrow (x^2 + y^2)(1 + m^2) = (a^2 + b^2)(1 + m^2)$$

That is, $x^2 + y^2 = a^2 + b^2$ is the required locus.

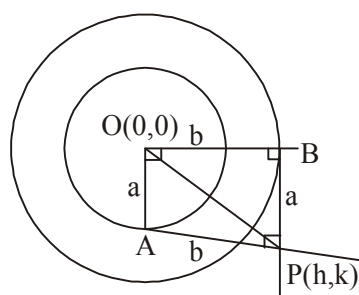
Note : in the problem if $b = a$, then the equation of the director circle of $x^2 + y^2 = a^2$ is $x^2 + y^2 = 2a^2$.

Alter :

$$OP = \sqrt{a^2 + b^2}$$

$$\sqrt{h^2 + k^2} = \sqrt{a^2 + b^2}$$

$$x^2 + y^2 = a^2 + b^2$$



- 18 If the variable line $3x - 4y + k = 0$ lies between the circles $x^2 + y^2 - 2x - 2y + 1 = 0$ and $x^2 + y^2 - 16x - 2y + 61 = 0$ without intersecting or touching either circle, then the range of k is (a, b) where $a, b \in I$. Find the value of $(b - a)$.

यदि चर रेखा $3x - 4y + k = 0$ वृत्तों $x^2 + y^2 - 2x - 2y + 1 = 0$ तथा $x^2 + y^2 - 16x - 2y + 61 = 0$ के बीच में बिना किसी वृत्त को स्पर्श या प्रतिच्छेद करते हुये स्थित हो, तो यदि k का परिसर (a, b) है जहाँ $a, b \in I$ तो $(b - a)$ का मान ज्ञात कीजिए।

Ans. 6

Sol. The given circles :

$$c_1(x - 1)^2 + (y - 1)^2 = 1$$

$$c_2(x - 8)^2 + (y - 1)^2 = 4$$

$$3x - 4y + k = 0$$

$$(3 \cdot 1 - 4 \cdot 1 + k)(3 \cdot 8 - 4 \cdot 1 + k) < 0 \Rightarrow (k - 1)(k + 20) < 0 \Rightarrow k \in (-20, 1)$$

For C_1

$$\frac{|3 \cdot 1 - 4 \cdot 1 + k|}{5} > 1 \Rightarrow \frac{|k - 1|}{5} > 1$$

$$\Rightarrow k - 1 > 5 \text{ or } k - 1 < -5$$

$$\Rightarrow k > 6 \text{ or } k < -4$$

For C_2

$$\frac{|3 \cdot 8 - 4 \cdot 1 + k|}{5} > 2 \Rightarrow \frac{|k + 20|}{5} > 2$$

$$\Rightarrow k + 20 > 10 \text{ or } k + 20 < -10$$

$$\Rightarrow k > -10 \text{ or } k < -30$$

$$\Rightarrow k \Rightarrow (-10, -4) \Rightarrow a = -10, b = -4$$

$$\Rightarrow b - a = -4 + 10 = 6$$

- 19 A circle with center in the first quadrant is tangent to $y = x + 10$, $y = x - 6$, and the y -axis. Let (h, k) be the center of the circle. If the value of $(h + k) = a + b\sqrt{a}$, find the value of $a + b$.

एक वृत्त का केन्द्र प्रथम चतुर्थांश में है तथा यह वृत्त रेखाओं $y = x + 10$, $y = x - 6$ तथा y -अक्ष को स्पर्श करता है। यदि (h, k) वृत्त का केन्द्र है तथा $(h + k)$ का मान $a + b\sqrt{a}$ है, $a + b$ का मान ज्ञात कीजिए।

Ans. 10

Sol. $y = x + 10$ and $y = x - 6$ have same slope, i.e. m for these lines to be the tangent to the circle, the perpendicular distance between them should be equal to the diameter of the circle.

i.e., $y = x + 2$ passes through the centre of the circle.

$$\Rightarrow k = h + 2$$

$$\text{So, radius of the circle} = \frac{1}{2} \left| \frac{16}{\sqrt{2}} \right| = 4\sqrt{2}$$

As y -axis is a tangent to the circle, $h = 4\sqrt{2}$, i.e. $k = 2 + 4\sqrt{2}$ So, $a + b = 2 + 8 = 10$

POWER OF POINT

20. A point moving around circle $(x + 4)^2 + (y + 2)^2 = 25$ with centre C broke away from it either at the point A or point B on the circle and moved along a tangent to the circle passing through the point $D(3, -3)$. Find the following.
- Equation of the tangents at A and B .
 - Coordinates of the points A and B .
 - Angle ADB and the maximum and minimum distances of the point D from the circle.
 - Area of quadrilateral $ADBC$ and the $\triangle DAB$.
 - Equation of the circle circumscribing the $\triangle DAB$ and also the intercepts made by this circle on the coordinate axes.

एक बिन्दु जो वृत्त $(x + 4)^2 + (y + 2)^2 = 25$ पर विचरण कर रहा है, वह वृत्त से बिन्दु A या बिन्दु B पर अलग हो जाता है तथा स्पर्श रेखा पर विचरण करता हुआ बिन्दु D $(3, -3)$ से गुजरता है। निम्न ज्ञात कीजिए।

- व B पर स्पर्श रेखाओं का समीकरण
- व B के निर्देशांक
- कोण ADB व D की वृत्त से अधिकतम व न्यूनतम दूरी
- चतुर्भुज ADBC व त्रिभुज $\triangle DAB$ का क्षेत्रफल
- त्रिभुज $\triangle DAB$ के परिवृत्त का समीकरण तथा इस वृत्त के द्वारा अक्षों पर काटे गये अन्तः खण्ड

Ans. (i) $3x - 4y = 21$; $4x + 3y = 3$ (ii) A(0, 1) and B $(-1, -6)$ (iii) 90° , $5(\sqrt{2} \pm 1)$ units
(iv) 25 sq. units, 12.5 sq. units (v) $x^2 + y^2 + x + 5y - 6 = 0$, x intercept 5; y intercept 7

Sol. (i) $CD = \sqrt{45+1} = \sqrt{50}$

$$AD = \sqrt{CD^2 + AC^2}$$

$$= \sqrt{50 - 25} = 5$$

Hence $AC = BC = BD = AD = 5$

$$\angle ADC = \angle BDC = 45^\circ$$

Let

$$\text{Slope of AD} = m_1$$

$$\text{Slope of BD} = m_2$$

$$\text{Slope of CD} = \frac{-2+3}{-4-3} = \frac{-1}{7}$$

$$\frac{\pi}{4} = \left| \frac{m + \frac{1}{7}}{1 - \frac{m}{7}} \right|$$

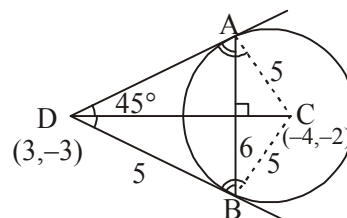
$$\frac{\pi}{4} = \left| \frac{m + \frac{1}{7}}{1 - \frac{m}{7}} \right|$$

Taking +ve

$$\left(1 - \frac{m}{7}\right) = \left(\frac{1}{7} + m\right)$$

Taking -ve

$$-1 + \frac{m}{7} = m + \frac{1}{7}$$



$$m = \frac{3}{4}$$

Equation of line

$$y + 3 = \frac{3}{4}(x - 3)$$

$$4y - 3x + 21 = 0$$

$$(ii) \text{ Slope of DC} = \frac{-1}{\text{slope of AD}}$$

$$= \frac{-4}{3}$$

Equation of AC

$$y + 2 = \frac{-4}{3}(x + 4)$$

$$3y + 6 = -4x - 16$$

$$4x + 3y + 22 = 0$$

For point A, intersection of AC and AD

For point B, intersection of BC and BD

Point A = (0, 1)

B = (-1, -6)

$$(iii) \text{ Minimum distance} = 5\sqrt{2} - 5$$

$$\text{Maximum distance} = 5\sqrt{2} + 5$$

$$(iv) \text{ Area of quadrilateral ABBC} = 5^2 = 25$$

$$\text{Area of ODAB} = \frac{1}{2} \times 25 = 12.5$$

$$(v) \text{ Mid point of CD} = \left(\frac{-1}{2}, \frac{-5}{2} \right)$$

Equation of circle,

$$\left(x + \frac{1}{2} \right)^2 + \left(y + \frac{5}{2} \right)^2 = \left(\frac{5\sqrt{2}}{2} \right)^2$$

$$x^2 + y^2 + x + 5y - 6 = 0$$

$$m = \frac{-4}{3}$$

Equation of line

$$y + 3 = \frac{-4}{3}(x - 3)$$

$$3y + 4x - 3 = 0$$

Slope of BC =

$$= \frac{3}{4}$$

21. Through a given point $P(5, 2)$, secants are drawn to cut the circle $x^2 + y^2 = 25$ at points $A_1(B_1)$, $A_2(B_2)$, $A_3(B_3)$, $A_4(B_4)$ and $A_5(B_5)$ such that $PA_1 + PB_1 = 5$, $PA_2 + PB_2 = 6$, $PA_3 + PB_3 = 7$, $PA_4 + PB_4 = 8$ and $PA_5 + PB_5 = 9$.

Find the value of $\sum_{i=1}^5 PA_i^2 + \sum_{i=1}^5 PB_i^2$. [Note : $A_r(B_r)$ denotes that the line passing through $P(5, 2)$ meets the circle $x^2 + y^2 = 25$ at two points A_r and B_r .]

यदि दिये गये एक बिन्दु $P(5, 2)$ से वृत्त $x^2 + y^2 = 25$ पर डाली गयी छेदक रेखायें वृत्त को बिन्दुओं $A_1(B_1)$, $A_2(B_2)$, $A_3(B_3)$, $A_4(B_4)$ एवं $A_5(B_5)$ पर प्रतिच्छेद करती है कि $PA_1 + PB_1 = 5$, $PA_2 + PB_2 = 6$, $PA_3 + PB_3 = 7$, $PA_4 + PB_4 = 8$ तथा $PA_5 + PB_5 = 9$ है।

$\sum_{i=1}^5 PA_i^2 + \sum_{i=1}^5 PB_i^2$ का मान ज्ञात कीजिए। [नोट : $A_r(B_r)$ का तात्पर्य वह रेखा है जो बिन्दु $P(5, 2)$ से गुजरती है जो वृत्त $x^2 + y^2 = 25$ को दो बिन्दुओं A_r तथा B_r पर काटती है।]

Ans. 215

Sol. $A^2 + B^2 = (A + B)^2 - 2AB$

$$\Rightarrow PA_1^2 + PA_2^2 + PA_3^2 + PA_4^2 + PA_5^2 + PB_1^2 + PB_2^2 + PB_3^2 + PB_4^2 + PB_5^2$$

$$\Rightarrow (PA_1 + PB_1)^2 - 2PA_1PB_1 + (PA_2 + PB_2)^2 - 2PA_2PB_2 + (PA_3 + PB_3)^2 - 2PA_3PB_3 + (PA_4 + PB_4)^2 - 2PA_4PB_4 + (PA_5 + PB_5)^2 - 2PA_5PB_5$$

$$\Rightarrow 25 + 36 + 49 + 64 + 81 - (2 \times 5 PT^2) \Rightarrow \text{Since we know } [PA \cdot PB = PT^2]$$

$$\Rightarrow 255 - 10PT^2 \quad \dots(1)$$

$$\text{Also, using distance formula } \Rightarrow PT = \sqrt{(5-5)^2 + (2-0)^2}$$

$$\Rightarrow PT = 2$$

$$\Rightarrow PT^2 = 4 \quad \dots(2)$$

Using (2) and (1)

$$\Rightarrow 255 - 10 \times 4 = 215$$

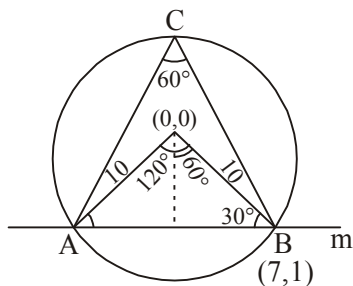
DIRECTOR CIRCLE

22. Chord AB of the circle $x^2 + y^2 = 100$ passes through the point $(7, 1)$ and subtends an angle of 60° at the circumference of the circle. If m_1 and m_2 are the slopes of two such chords then the value of $m_1 m_2$, is :

वृत्त $x^2 + y^2 = 100$ की जीवा AB बिन्दु $(7, 1)$ से जाती है तथा परिधि के साथ 60° का कोण बनाती है। यदि m_1 तथा m_2 ऐसी जीवाओं के ढलान हैं तो $m_1 m_2$ का मान है :

Ans. -1

Sol.



CHORD WHOSE MID POINT IS GIVEN

23. Find the locus of the mid point of the chord of a circle $x^2 + y^2 = 4$ such that the segment intercepted by the chord on the curve $x^2 - 2x - 2y = 0$ subtends a right angle at the origin.

वृत्त $x^2 + y^2 = 4$ की उस जीवा के मध्य बिन्दु का बिन्दुपथ ज्ञात करें जिस जीवा के द्वारा वक्र $x^2 - 2x - 2y = 0$ पर काटा गया अन्तःखण्ड मूल बिन्दु पर समकोण बनाता है।

Ans. $x^2 + y^2 - 2x - 2y = 0$

Sol. Let the midpoint be (h, k)

Equation of chord $T = S$

$$\Rightarrow hx + ky = h^2 + k^2$$

Let the above line intersects curve $x^2 - 2x - 2y = 0$ at A and B.

Let O be the origin.

For finding $OA \times OB$, we will use homogenization

$$x^2 - 2x \left(\frac{hx + ky}{h^2 + k^2} \right) - 2y \left(\frac{hx + ky}{h^2 + k^2} \right) = 0$$

$$x^2 (h^2 + k^2) - x(hx + ky) - 2y(hx + ky) = 0$$

$$\Rightarrow x^2 (h^2 + k^2) + x^2 (-2h) - 2kxy - 2hxy - 2ky^2 = 0$$

$$\Rightarrow x^2 (h^2 + k^2 - 2h) + y^2 (-2k) + y(-2hx - 2kx) = 0$$

$\therefore$ OA and OB are perpendicular.

$$\Rightarrow h^2 + k^2 - 2h - 2k = 0$$

$$\Rightarrow h^2 + k^2 = 2(h + k)$$

$\therefore$ Locus is $x^2 + y^2 = 2(x + y)$

$$\Rightarrow x^2 + y^2 - 2x - 2y = 0$$

24. Find the locus of the mid point of all chords of the circle $x^2 + y^2 - 2x - 2y = 0$ such that the pair of lines joining $(0, 0)$ and the point of intersection of the chords with the circles make equal angle with axis of x .

वृत्त $x^2 + y^2 - 2x - 2y = 0$ की उन जीवाओं के मध्य बिन्दु का बिन्दुपथ ज्ञात करें जिनके वृत्त के साथ प्रतिच्छेद बिन्दुओं को $(0, 0)$ से मिलाने वाली रेखा युग्म x अक्ष के साथ समान कोण बनाता है।

Ans. $x + y = 2$

Sol. Let the midpoint of chord be (h, k) .

Equation of chord $T = S_1$

$$\Rightarrow hx + ky - (x + h) - (y + k) = h^2 + k^2 - 2h - 2k$$

Homogenising equation of circle with the above chord equation gives $OA \times OB$ equation.

$$x^2 + y^2 - 2(x + y) \left(\frac{hx + ky - x - y}{h^2 + k^2 - h - k} \right) = 0$$

$$\Rightarrow x^2 (h^2 + k^2 - h - k) + y^2 (h^2 + k^2 - h - k) - 2(hx^2 + kxy - x^2 - xy + hxy + ky^2 - xy - y^2) = 0$$

$$\Rightarrow x^2 (h^2 + k^2 - h - k - 2h + 2) + y^2 (h^2 + k^2 - h - k - 2k + 2) + xy (-2k + 2 - 2h + 2) = 0$$

$$\Rightarrow x^2 (h^2 + k^2 - h - k - 2h + 2) + y^2 (h^2 + k^2 - h - k - 2k + 2) + 2xy (-k - h + 2) = 0$$

$$\Rightarrow \text{It represents the form } ax^2 + by^2 + 2hxy \text{ and } m_1 + m_2 = \frac{-2h}{b}$$

Where m_1 and m_2 are slopes of two lines represented by the equation.

$\therefore$ OA and OB make equal angle with x -axis.

$$\Rightarrow \frac{-2h}{b} = 0$$

$$\Rightarrow \frac{-2(2 - k - h)}{h^2 + k^2 + 2 - h - 3k} = 0$$

$$\Rightarrow 2 - k - h = 0$$

$$\Rightarrow h + k = 2$$

$$\therefore \text{Locus : } x + y = 2$$

CHORD OF CONTACT

25. Find the equation of the circle passing through the three points $(4, 7)$, $(5, 6)$ and $(1, 8)$. Also find the coordinates of the point of intersection of the tangents to the circle at the points where it is cut by the straight line $5x + y + 17 = 0$.

तीन बिन्दुओं $(4, 7)$, $(5, 6)$ व $(1, 8)$ से गुजरने वाले वृत्त का समीकरण ज्ञात करें। उन बिन्दुओं पर खींची गई स्पर्श रेखाओं का प्रतिच्छेद बिन्दु ज्ञात करें जहाँ $5x + y + 17 = 0$ वृत्त को प्रतिच्छेद करती है।

Ans. $(-4, 2), x^2 + y^2 - 2x - 6y - 15 = 0$

Sol. Equation : $(x-h)^2 + (y-k)^2 = r^2$

A : $(4-h)^2 + (7-k)^2 = r^2$

$\Rightarrow 16 + h^2 - 8h + 49 + k^2 - 14k = r^2$

$\Rightarrow h^2 + k^2 - 8h - 14k + 65 = r^2$ _____(1)

B : $(5-h)^2 + (6-k)^2 = r^2$

$\Rightarrow h^2 + k^2 - 10h - 12k + 61 = r^2$ _____(2)

C : $(1-h)^2 + (8-k)^2 = r^2$

$h^2 + k^2 - 2h - 16k + 65 = r^2$ _____(3)

Subtracting equation (1) and (2)

$h^2 + k^2 - 8h - 14k + 65 = r^2$

$h^2 + k^2 - 2h - 16k + 65 = r^2$

$-6k + 2k = 0 \dots (4) \times 2$

$64 = 2k$

$-12h + 12k = 0$ _____(5)

Subtracting equation (2) and (3)

$h^2 + k^2 - 2h - 16k + 65 = r^2$

$h^2 + k^2 - 10h - 12k + 61 = r^2$

$8h - 4k + 4 = 0$ _____(6)

Adding equation (5) and (6)

$-12h + 4k = 0$

$8h - 4k + 4 = 0$

$-4h + 4 = 0$

$h = 1$

$-12 + 4k = 0$

$4k = 12$

$k = 3$

$\therefore \text{center } O \equiv (h, k) = (1, 3)$

Radius : $OA = \sqrt{(1-4)^2 + (3-7)^2} = \sqrt{3^2 + 4^2}$

$$\sqrt{9+16} = \sqrt{25} = 5 \text{ cm}$$

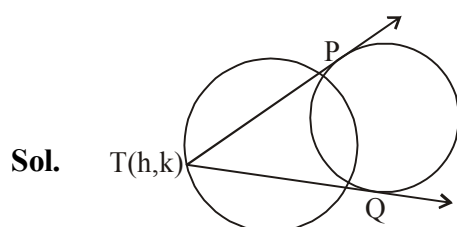
$$\text{Equation of circle : } (x - h)^2 + (y - k)^2 = r^2$$

$$\Rightarrow (x - 1)^2 + (y - 3)^2 = 5^2$$

26. Tangents TP and TQ are drawn from a point T to the circle $x^2 + y^2 = a^2$. If the point T lies on the line $px + qy = r$, find the locus of centre of the circumcircle of triangle TPQ.

बिन्दु T से वृत्त $x^2 + y^2 = a^2$ पर TP व TQ स्पर्श रेखायें खींची गई हैं। यदि T रेखा $px + qy = r$ पर स्थित है, तो त्रिभुज TPQ के परिवृत्त के केन्द्र का बिन्दुपथ ज्ञात कीजिए।

Ans. $2px + 2qy = r$



Circle TPQ is $S + \lambda P = 0$ where P is chord of contact of T(h, k).

$$\text{Circumcircle is } (x^2 + y^2 - a^2) + \lambda(hx + ky - a^2)$$

It passes through T(h, k)

$$\therefore (h^2 + k^2 - a^2) + \lambda(h^2 + k^2 - a^2) = 0$$

$$\text{or } \lambda = -1$$

$$\text{Circle is } x^2 + y^2 - 4x - ky = 0$$

Its centre $x = \frac{h}{2}, y = \frac{k}{2}$. But (h, k) lies on $px + qy + r = 0$, $\therefore ph + qk + r = 0$.

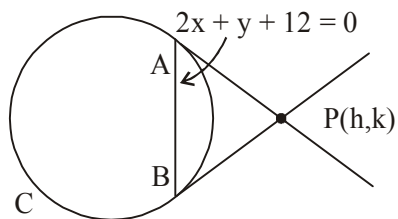
Eliminating h, k we get the locus as $p \cdot 2x + q \cdot 2y + r = 0$.

27. Find the point of intersection of the tangents drawn to the circle $x^2 + y^2 - 4x + 3y - 1 = 0$ at the points where the line $2x + y + 12 = 0$ meets it.

रेखा $2x + y + 12 = 0$ द्वारा वृत्त $x^2 + y^2 - 4x + 3y - 1 = 0$ काटे गए बिन्दुओं पर स्पर्श रेखाओं का प्रतिच्छेदन बिन्दु ज्ञात करें।

Ans. (1, -2)

Sol.



$$S : x^2 + y^2 - 4x + 3y - 1 = 0.$$

Let the point of intersection of tangents by $p(h, k)$.

Line aB $2x + y + 12 = 0$ is chord of contact.

Equation of chord of contact is given by :

$$x(h) + y(k) - 4\left(\frac{x+h}{2}\right) + 3\left(\frac{y+k}{2}\right) - 1 = 0$$

$$\Rightarrow xh + yk - 2x - 2h + \frac{3y}{2} + \frac{3k}{2} - 1 = 0$$

$$\Rightarrow x(h-2) + y\left(k + \frac{3}{2}\right) - \left(2h - \frac{3k}{2} + 1\right) = 0$$

Comparing this with given equation $2x + y + 12 = 0$

$$\frac{h-2}{2} = \frac{2k+3}{2} = -\left(\frac{4h-3k+2}{24}\right)$$

Solving, we get

$$h = 1, k = -2$$

$$(1, -2).$$

POLE AND POLAR

28. The chords of contact of the pair of tangents to the circle $x^2 + y^2 = 1$ drawn from any point on the line $2x + y = 4$ pass through the point (α, β) , then find $\alpha^2 + \beta^2$.

रेखा $2x + y = 4$ के किसी बिन्दु से वृत्त $x^2 + y^2 = 1$ के सापेक्ष सभी स्पर्श रेखाएँ बिन्दु (α, β) से गुजरती है, तो $\alpha^2 + \beta^2$ का मान ज्ञात कीजिए।

Ans. $\frac{5}{16}$

Sol. Let (h, k) be any point on the given line

$$\therefore 2h + k = 4 \text{ and C.C. is } hx + ky = 1$$

$$\text{or } hx + (4 - 2h)y = 1 \quad \therefore k = 4 - 2h$$

$$\text{or } (4y - 1) + h(x + 2y) = 0$$

It is of the form $P + \lambda Q = 0$. It passes through the intersection of $P = 0$ and $Q = 0$ or $\left(\frac{1}{2}, \frac{1}{4}\right)$.

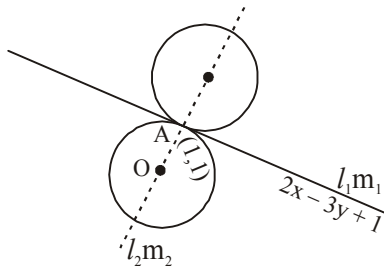
FAMILY OF CIRCLES

29. The line $2x - 3y + 1 = 0$ is tangent to a circle $S = 0$ at $(1, 1)$. If the radius of the circle is $\sqrt{13}$. Find the equation of the circle S .

रेखा $2x - 3y + 1 = 0$ वृत्त $S = 0$ को $(1, 1)$ पर स्पर्श करती है। यदि वृत्त की त्रिज्या $\sqrt{13}$ है तो वृत्त का समीकरण ज्ञात कीजिए।

Ans. $x^2 + y^2 - 6x + 4y = 0$ OR $x^2 + y^2 + 2x - 8y + 4 = 0$

Sol.



$$\text{Slope of line } \ell_2 = \frac{-1}{\text{slope of line } \ell_1}$$

$$= -\frac{1}{\frac{2}{3}}$$

$$= -\frac{3}{2}$$

$$\text{Equation of line } \ell_2 = y - 1 = -\frac{3}{2}(x - 1)$$

$$y = -\frac{3}{2}x + \frac{5}{2}$$

$$\text{Let coordinate of centre} \equiv \left(a, -\frac{3a}{2} + \frac{5}{2}\right)$$

$$\text{Thus } OA = \sqrt{13}$$

$$OA^2 = 13$$

$$(a-1)^2 + \left(-\frac{3a}{2} + \frac{5}{2} - 1\right)^2 = 13$$

$$(a-1)^2 + \frac{9}{4}(a-1)^2 = 13$$

$$(a-1)^2 \times \frac{13}{4} = 13$$

$$(a-1)^2 = 4$$

$$a = 3, -1$$

Thus coordinates of centre are (3, -2) & (-1, 4)

Equation of circles

$$(x-3)^2 + (y+2)^2 = 13 \text{ \& } (x+1)^2 + (y-4)^2 = 13$$

Note : (1) Two circles come from the solution which is true because one circle on each side of line is possible.

(2) Distance Formula = $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$ between two points (x_1, y_1) & (x_2, y_2) .

(3) Putting x-coordinate which is a in line L_2 we get y-coordinate of centre.

30. Find the equation of the circle which passes through the point (1, 1) & which touches the circle $x^2 + y^2 + 4x - 6y - 3 = 0$ at the point (2, 3) on it.

उस वृत्त का समीकरण ज्ञात करें जो (1, 1) से गुजरता है तथा वृत्त $x^2 + y^2 + 4x - 6y - 3 = 0$ को (2, 3) पर स्पर्श करता है।

Ans. $x^2 + y^2 + x - 6y + 3 = 0$

Sol. Equation of tangent to circle

$$x^2 + y^2 + 4x - 6y - 3 = 0 \text{ at } (2, 3) \Rightarrow 2x + 3y + 2(x+2) - 3(y+3) - 3 = 0$$

$$4x - 8 = 0 \Rightarrow x - 2 = 0$$

$$\text{family of circle} \Rightarrow S + \lambda L = 0$$

$$x^2 + y^2 + 4x - 6y - 3 + \lambda(x - 2) = 0 \dots\dots\dots(1)$$

Passes through (1, 1)

$$-3 - \lambda = 0 \Rightarrow \lambda = -3$$

Putting in (1)

$$x^2 + y^2 + x - 6y + 3 = 0$$

31. Find the equation of the circle whose radius is 3 and which touches the circle $x^2 + y^2 - 4x - 6y - 12 = 0$ internally at the point $(-1, -1)$.

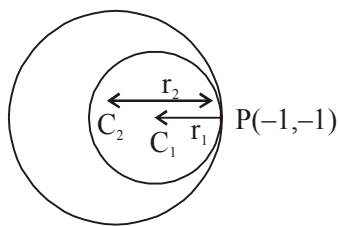
उस वृत्त का समीकरण ज्ञात करें जिसकी त्रिज्या 3 है तथा जो वृत्त $x^2 + y^2 - 4x - 6y - 12 = 0$ को बिन्दु $(-1, -1)$ पर आन्तरिक स्पर्श करता है।

Ans. $5x^2 + 5y^2 - 8x - 14y - 32 = 0$

Sol. $5x^2 + 5y^2 - 8x - 14y - 32 = 0$

$C_1 : (x_1, y_1), r_1 = 3$

$C_2 : (2, 3), r_2 = 5$



Point P divides C_2C_1 externally in the ratio of their radius.

$C_2P : C_1P = -5 : 3$

or $C_1C_2 : C_1P = 2 : 3$

co-ordinates of $C_1 = \left(\frac{4}{5}, \frac{7}{5}\right)$

Equation of circle is

$$\left(x - \frac{4}{5}\right)^2 + \left(y - \frac{7}{5}\right)^2 = 9$$

$$\Rightarrow 5x^2 + 5y^2 - 8x - 14y - 32 = 0$$

32. Find the equation of the circle passing through the points of intersection of the circles $x^2 + y^2 - 2x - 4y - 4 = 0$ and $x^2 + y^2 - 10x - 12y + 40 = 0$ and whose radius is 4.

वृत्तों $x^2 + y^2 - 2x - 4y - 4 = 0$ एवं $x^2 + y^2 - 10x - 12y + 40 = 0$ के प्रतिच्छेदन बिन्दुओं से गुजरने वाले वृत्त का समीकरण ज्ञात करें जिसकी त्रिज्या 4 है।

Ans. $2x^2 + 2y^2 - 18x - 22y + 69 = 0$ and $x^2 + y^2 - 2y - 15 = 0$

Sol. Equation of the any circle through the points of intersection of given circles is

$$(x^2 + y^2 - 2x - 4y - 4) + \lambda (x^2 + y^2 - 10x - 12y + 40) = 0$$

$$\Rightarrow x^2(1 + \lambda) + y^2(1 + \lambda) - 2x(1 + 5\lambda) - 2y(2 + 6\lambda) - 4 + 40\lambda = 0$$

$$\text{or } x^2 + y^2 - 2x \frac{(1+5\lambda)}{(1+\lambda)} - 2y \frac{(2+6\lambda)}{(1+\lambda)} + \frac{(40\lambda-4)}{(1+\lambda)} = 0 \quad \text{_____ (1)}$$

$$\text{Its radius } \sqrt{\left(\frac{1+5\lambda}{1+\lambda}\right)^2 + \left(\frac{2+6\lambda}{1+\lambda}\right)^2 - \left(\frac{40\lambda-4}{1+\lambda}\right)} = 4 \quad (\text{given})$$

$$\Rightarrow \frac{(1+5\lambda)^2 + (2+6\lambda)^2 - (40\lambda-4)(1+\lambda)}{(1+\lambda)^2} = 16$$

$$\Rightarrow 5\lambda^2 + 34\lambda - 7 = 0$$

$$\text{or } (\lambda - 7)(5\lambda + 1) = 0$$

$$\therefore \lambda = 7 \text{ or } \lambda = -\frac{1}{5}$$

Substituting the values of λ in (1), the required circles are

$$2x^2 + 2y^2 - 18x - 22y + 69 = 0 \text{ and } x^2 + y^2 - 2y - 15 = 0 \text{ Ans.}$$

33. Find the equation to the circle which passes through the points $(1, -2)$ and $(4, -3)$ and which has its centre on the straight line $3x + 4y = 7$.

बिन्दु $(1, -2)$ एवं $(4, -3)$ से गुजरने वाले वृत्त का समीकरण ज्ञात कीजिये, जिसका केन्द्र सरल रेखा $3x + 4y = 7$ पर स्थित है।

Ans. $15x^2 + 15y^2 - 94x + 18y + 55 = 0$

Sol. Let the equation of circle be $x^2 + y^2 + 2gx + 2fy + c = 0$

Centre of circle is $(-g, -f)$.

Centre lies on the line $3x + 4y = 7$.

Therefore,

$$-3g - 4f = 7$$

$$3g + 4f + 7 = 0 \quad \text{_____ (1)}$$

Circle passes through the points $(1, -2)$ and $(4, -3)$.

By putting these values in the equation of circle,

$$2g - 4f + c + 5 = 0 \text{ and } 8g - 6f + c + 25 = 0$$

Solving these two equation we get,

$$6g - 2f + 20 = 0 \quad \text{_____ (2)}$$

Now, solving equation (1) & (2), we get,

$$f = \frac{3}{5} \text{ and } g = \frac{-47}{15}$$

Also, $c = \frac{11}{3}$

Hence, the equation of circle is,

$$x^2 + y^2 + 2x\left(\frac{-47}{15}\right) + 2y\left(\frac{3}{5}\right) + \frac{11}{3}$$

$$15(x^2 + y^2) - 94x + 18y + 55 = 0$$

POSITION OF TWO CIRCLES AND COMMON TANGENTS BETWEEN THEM

34. Let P and Q be 2 circles externally tangent at point X. A straight line is tangent to P at point A and is tangent to Q at point B ($A \neq B$). The line tangent to P and Q at X intersects line AB at a point Y. If $AY = 10$ and the radius of P is 9. If the radius of the circle Q can be expressed as a rational number a/b in the lowest form, find the value of $(a + b)$.

माना P तथा Q, दो वृत्त जो बिन्दु X पर बाह्य स्पर्श करते हैं। एक सरल रेखा P को बिन्दु A पर तथा Q को बिन्दु B पर ($A \neq B$) स्पर्श करती है। बिन्दु X पर खींची गई स्पर्श रेखा AB को Y पर प्रतिच्छेद करती है। $AY = 10$ है तथा P की त्रिज्या 9 है। यदि वृत्त Q की त्रिज्या को परिमेय संख्या a/b के न्यूनतम समकक्ष रूप में प्रदर्शित किया जा सकता है तो $(a + b)$ का मान ज्ञात कीजिये।

Ans. 109

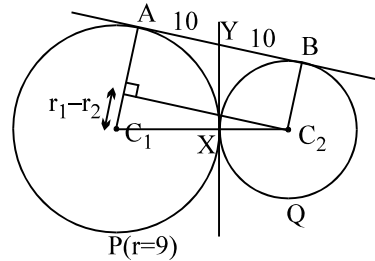
Sol. $AY = BY$ ($\because$ Y lies on radical axis)

$$(AB)^2 = (r_1 + r_2)^2 - (r_1 - r_2)^2 = 4r_1r_2$$

$$AB = 2\sqrt{r_1r_2}$$

$$AB = 20 = 2\sqrt{9r_2}$$

$$r_2 = \frac{100}{9} \text{ Ans.}$$



35. Find the sum of all integral value(s) of 'r' ($r > 0$) for which the circles $x^2 + y^2 - 10x + 16y + 89 - r^2 = 0$ and $x^2 + y^2 + 6x - 14y + 42 = 0$ intersect in two real distinct points.

यदि वृत्त $x^2 + y^2 - 10x + 16y + 89 - r^2 = 0$ तथा $x^2 + y^2 + 6x - 14y + 42 = 0$ एक दूसरे को यदि दो भिन्न-भिन्न बिन्दुओं पर काटते हैं 'r' के सभी सम्भावित ($r > 0$) पूर्णांक मानों का योग क्या होगा।

Ans. 119

Sol. We have $(x - 5)^2 + (y + 8)^2 = 25 + 64 + r^2 - 89$

$$\text{and } (x + 3)^2 + (y - 7)^2 = 49 + 9 - 42 = 16$$

$$\Rightarrow (x - 5)^2 + (y + 8)^2 = r^2$$

$$\text{and } (x + 3)^2 + (y - 7)^2 = (4)^2$$

$$\frac{(5,-8) \quad (-3,7)}{\Rightarrow \sqrt{64+225} = \sqrt{289} = 17 = \text{distance between their centres}}$$

$$\text{Now, } |r-4| < 17 < r+4 \Rightarrow r+4 > 17 \Rightarrow r > 13$$

$$\text{and } -17 < r-4 < 17 \Rightarrow -13 < r < 21$$

$$\text{Hence } 13 < r < 21$$

$\therefore$ Possible values of 'r' can be 14, 15, 16, 17, 18, 19, 20

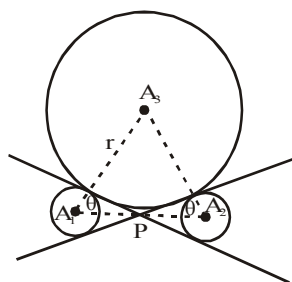
Hence required sum = 119

36. The centres of two circles C_1 and C_2 each of unit radius are at a distance of 6 units from each other. Let P be the mid point of the line segment joining the centres of C_1 and C_2 and C be a circle touching circles C_1 and C_2 externally. If a common tangent to C_1 and C passing through P is also a common tangent to C_2 and C, then the radius of the circle C is. [JEE-2009]

दो वृत्त जिनमें से प्रत्येक की त्रिज्या इकाई है, के केन्द्र परस्पर 6 इकाई दूर हैं। माना कि C_1 व C_2 को जोड़ने वाले रेखा खण्ड का मध्य बिन्दु P है तथा C_1 व C_2 की बाह्य स्पर्श करने वाला एक वृत्त C है। यदि C_1 व C_2 की एक उभयनिष्ठ स्पर्श रेखा जो P से गुजरती है, C_2 व C की भी उभयनिष्ठ स्पर्श रेखा है तो वृत्त C की त्रिज्या ज्ञात कीजिए। [JEE-2009]

Ans. 8

Sol.



In triangle $A_1A_2A_3$

$$A_1A_3 = A_3A_2$$

$$\text{Let angle } A_3A_1A_2 = \theta, \cos \theta = \frac{1}{3}, \sin \theta = \frac{2\sqrt{2}}{3}$$

Apply sine rule in triangle $A_1A_2A_3$

$$\frac{6}{\sin(\pi - 2\theta)} = \frac{r+1}{\sin \theta}$$

$$\Rightarrow r = 8$$

RADICAL AXIS AND ITS PROPERTIES

37. Show that the equation of a straight line meeting the circle $x^2 + y^2 = a^2$ in two points at equal distances 'd' from a point (x_1, y_1) on its circumference is $xx_1 + yy_1 - a^2 + \left(\frac{d^2}{2}\right) = 0$.

प्रदर्शित किजिये कि वृत्त $x^2 + y^2 = a^2$ पर स्थित बिन्दु (x_1, y_1) से समान दूरी 'd' पर स्थित दो बिन्दु, जो वृत्त $x^2 + y^2 = a^2$ पर स्थित

है, से गुजरने वाली सरल रेखा का समीकरण $xx_1 + yy_1 - a^2 + \left(\frac{d^2}{2}\right) = 0$ है।

Ans.

Sol. The equation of the circle is $x^2 + y^2 = a^2$

Let the points the line meets circle be,

A $(a \cos \theta_1, a \sin \theta_1)$ and $(a \cos \theta_2, a \sin \theta_2)$

Given these points are at distance 'd' from (x_1, y_1)

$$\Rightarrow (x_1 - a \cos \theta_1)^2 + (y_1 - a \sin \theta_1)^2 = (x_1 - a \cos \theta_2)^2 + (y_1 - a \sin \theta_2)^2 = d^2 \quad \dots(1)$$

$$\Rightarrow x_1^2 - 2ax_1 \cos \theta_1 + a^2 \cos^2 \theta_1 + y_1^2 + a^2 \sin^2 \theta_1 - 2ay_1 \sin \theta_1 = x_1^2 - 2ax_1 \cos \theta_2 + a^2 \cos^2 \theta_2 + y_1^2$$

$$-2ay_1 \sin \theta_2 + a^2 \sin^2 \theta_2$$

$$x_1 \cos \theta_1 + y_1 \sin \theta_1 = y_1 \sin \theta_2 + x_1 \cos \theta_2$$

$$\Rightarrow \frac{\sin \theta_2 - \sin \theta_1}{\cos \theta_2 - \cos \theta_1} = \frac{-x_1}{y_1} \quad \dots(2)$$

The equation of the line joining A and B is

$$(y - a \sin \theta_2) = \frac{a \sin \theta_2 - a \sin \theta_1}{a \cos \theta_2 - a \cos \theta_1} (x - a \cos \theta_2)$$

$$\Rightarrow (y - a \sin \theta_2) = \frac{-x_1}{y_1} (x - a \cos \theta_2)$$

$$\Rightarrow y_1 y + x x_1 - y_1 a \sin \theta_2 - a x_1 \cos \theta_2 = 0$$

$$\Rightarrow x x_1 + y y_1 + y y_1 + \left(\frac{x_1^2 - 2a x_1 \cos \theta_2 + a^2 \cos^2 \theta_2}{2} \right) + \left(\frac{y_1^2 - 2a y_1 \sin \theta_2 + a^2 \sin^2 \theta_2}{2} \right)$$

$$- \left(\frac{x_1^2 + y_1^2}{2} \right) - \frac{a^2}{2} = 0$$

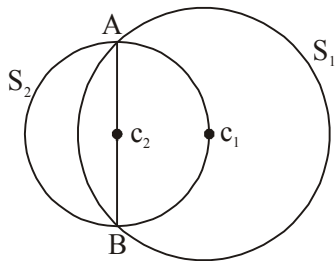
$$\Rightarrow x x_1 + y y_1 + \frac{d^2}{2} - a^2 = 0$$

38. If the circle $x^2 + y^2 + 4x + 22y + a = 0$ bisects the circumference of the circle $x^2 + y^2 - 2x + 8y - b = 0$ (where $a, b > 0$), then find the maximum value of (ab) .

यदि वृत्त $x^2 + y^2 + 4x + 22y + a = 0$, वृत्त $x^2 + y^2 - 2x + 8y - b = 0$ की परिधि को समद्विभाजित करता है (जहाँ $a, b > 0$), तो (ab) का अधिकतम मान ज्ञात कीजिये।

Ans. 625

Sol.



Given that $S_1 = x^2 + y^2 + 4x + 22y + c = 0$

$S_2 = x^2 + y^2 - 2x + 8y - d = 0$

AB is a common chord passing through C_2 .

Equation of AB : $S_1 - S_2 = 0$

$$\Rightarrow (x^2 + y^2 + 4x + 22y + c) - (x^2 + y^2 - 2x + 8y - d) = 0$$

$$\Rightarrow 6x + 14y + c + d = 0$$

The common chord passes through centre of S_2 , i.e. $C_2(1, -4)$

$$\Rightarrow 6(1) + 14(-4) + c + d = 0$$

$$\Rightarrow c + d = 50$$

Since A.M. $\geq$ G.M.

$$\frac{c+d}{2} \geq \sqrt{cd}$$

$$\Rightarrow 25 \geq \sqrt{cd}$$

$$\Rightarrow cd \leq 625$$

39. Find the equation of a circle which is co-axial with circles $2x^2 + 2y^2 - 2x + 6y - 3 = 0$ and $x^2 + y^2 + 4x + 2y + 1 = 0$. It is given that the centre of the circle to be determined lies on the radical axis of these two circles.

उस वृत्त का समीकरण ज्ञात करें जो वृत्त $2x^2 + 2y^2 - 2x + 6y - 3 = 0$ तथा $x^2 + y^2 + 4x + 2y + 1 = 0$ के साथ समाक्षी है। यह दिया गया है कि उस वृत्त का केन्द्र, जिसे ज्ञात करना है। इन दो वृत्तों के मूलाक्ष अक्ष पर स्थित है।

Ans. $4x^2 + 4y^2 + 6x + 10y - 1 = 0$

Sol. The two given circles in standard form we

$$S_1 = 0 \text{ or } x^2 + y^2 - x + 3y - \frac{3}{2} = 0$$

$$S_2 = 0 \text{ or } x^2 + y^2 + 4x + 2y + 1 = 0$$

Radical axis is given by $S_1 - S_2 = 0$

$$\therefore -5x + y - \frac{5}{2} = 0$$

$$\text{or } 10x - 2y + 5 = 0 \quad \dots(1)$$

Circle coaxial with the given circles is

$$S_1 + \lambda S_2 = 0$$

$$\left(x^2 + y^2 - x + 3y - \frac{3}{2} \right) + \lambda(x^2 + y^2 + 4x + 2y + 1) = 0$$

$$\text{or } (x^2 + y^2)(1 + \lambda) - (1 - 4\lambda)x + (3 + 2\lambda)y - \frac{3}{2} + \lambda = 0$$

$$\text{or } x^2 + y^2 - \frac{1-4\lambda}{1+\lambda}x + \frac{3+2\lambda}{1+\lambda}y + \frac{\lambda-\frac{3}{2}}{1+\lambda} = 0$$

$$\text{Its centre } \left(\frac{1}{2}, \frac{1-4\lambda}{1+\lambda}, -\frac{1}{2}, \frac{3+2\lambda}{1+\lambda} \right)$$

i.e. $(-g, -f)$ lies on (1)

$$\therefore \frac{5(1-4\lambda)}{1+\lambda} + \frac{3+2\lambda}{1+\lambda} + 5 = 0$$

$$\text{or } 5 - 20\lambda + 3 + 2\lambda + 5 + 5\lambda = 0$$

$$\text{or } 13 - 13\lambda = 0 \quad \therefore \lambda = 1$$

putting in (2), the required circle is

$$x^2 + y^2 + \frac{3}{2}x + \frac{5}{2}y - \frac{1}{4} = 0$$

$$\text{or } 4x^2 + 4y^2 + 6x + 10y - 1 = 0$$

ANGLE BETWEEN TWO CIRCLES & ORTHOGONALITY OF CIRCLES

40. Find the equation of the circle through the points of intersection of circles $x^2 + y^2 - 4x - 6y - 12 = 0$ and $x^2 + y^2 + 6x + 4y - 12 = 0$ and cutting the circle $x^2 + y^2 - 2x - 4 = 0$ orthogonally.

वृत्तों $x^2 + y^2 - 4x - 6y - 12 = 0$ तथा $x^2 + y^2 + 6x + 4y - 12 = 0$ के प्रतिच्छेद बिन्दुओं से जाने वाले तथा वृत्त $x^2 + y^2 - 2x - 4 = 0$ के लम्बकोणीय वृत्त का समीकरण ज्ञात कीजिए।

Ans. $x^2 + y^2 + 16x + 14y - 12 = 0$

Sol. Let, the equation of the circle through the intersection of given circle is,

$$x^2 + y^2 - 4x - 6y + 12 = 0 \quad \dots(1)$$

Here, the equation of radical axis for the circles

$$x^2 + y^2 - 4x - 6y + 12 = 0$$

and $x^2 + y^2 + 6x + 4y - 12 = 0$ is,

$$-10x - 10y = 0$$

So, equation (1) can be written as,

$$x^2 + y^2 - 4x - 6y - 12 + \lambda(-10x - 10y) = 0$$

$$\Rightarrow x^2 + y^2 - x(10\lambda + 4) - y(10\lambda + 6) - 12 = 0$$

It cuts the circle v orthogonally.

$$\therefore 2gg_1 + 2ff_1 = c + c_1$$

$$\Rightarrow (10\lambda + 4)(1) + (10\lambda + 6)(0) = -12 - 4$$

$$\Rightarrow \lambda = -2$$

Hence, the required equation of the circle is,

$$x^2 + y^2 - 4x - 6y - 12 - 2(-10x - 10y) = 0$$

$$\Rightarrow x^2 + y^2 + 16x + 14y - 12 = 0$$

41. The centre of the circle $S = 0$ lie on the line $2x - 2y + 9 = 0$ and $S = 0$ cuts orthogonally the circle $x^2 + y^2 = 4$. Show that circle $S = 0$ passes through two fixed points & find their coordinates.

वृत्त $S = 0$ का केन्द्र रेखा $2x - 2y + 9 = 0$ पर स्थित है तथा $S = 0$ वृत्त $x^2 + y^2 = 4$ के लम्बकोणीय है। प्रदर्शित करें कि वृत्त $S = 0$ दो अचर बिन्दुओं से गुजरता है तथा उनके निर्देशांक ज्ञात कीजिए।

Ans. $(-4, 4); (-1/2, 1/2)$

Sol. Let the circle be

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Its centre $(-g, -f)$ lies on $2x - 2y + 9 = 0$.

$$\therefore -2g + 2f + 9 = 0 \quad \dots(1)$$

IT cuts orthogonally the circle $x^2 + y^2 - 4 = 0$

$$\therefore 2g_1g_2 + 2f_1f_2 = c_1 + c_2$$

$$\text{or } 0 + 0 = c - 4 \quad \therefore c = 4$$

and from (1), $2g = 9 + 2f$.

Putting the values of c and g we get the equation of the circle as

$$x^2 + y^2 + (9 + 2g)x + 2fy + 4 = 0$$

$$\text{or } (x^2 + y^2 + 9x + 4) + 2f(x + y) = 0$$

Above is of the form $S + \lambda P = 0$

Above represents a family of circles which passes through the points of intersection of $S = 0$ and $P = 0$.

Solving $x^2 + y^2 + 9x + 4 = 0$ and $x + y = 0$ we get the fixed points as $\left(-\frac{1}{2}, \frac{1}{2}\right)$ and $(-4, 4)$.

42. Find the equation of a circle passing through the origin if the line pair, $xy - 3x + 2y - 6 = 0$ is orthogonal to it. If this circle is orthogonal to the circle $x^2 + y^2 - kx + 2ky - 8 = 0$ then find the value of k .

मूल बिन्दु से गुजरने वाले वृत्त का समीकरण ज्ञात करें, यदि रेखा युग्म $xy - 3x + 2y - 6 = 0$ वृत्त के लम्बकोणीय हो। यदि यह वृत्त, वृत्त $x^2 + y^2 - kx + 2ky - 8 = 0$ के लम्बकोणीय है तो k का मान ज्ञात कीजिए।

Ans. $x^2 + y^2 + 4x - 6y = 0$; $k = 1$

Sol. Let the equation of circle be

$$x^2 + y^2 + 2gx + 2fy = 0$$

it passes through origin)

The line pair is

$$xy - 3x + 2y - 6 = 0$$

$$x(y - 3) + 2(y - 3) = 0$$

$$(x + 2)(y - 3) = 0$$

The centre is point of intersection of these two lines.

$$c \equiv (-2, 3)$$

$$g = 2 \text{ and } f = -3$$

$$s \equiv x^2 + y^2 + 4x - 6y = 0$$

$$s_1 = x^2 + y^2 - kx + 2ky - 8 = 0$$

Since s & s_1 are orthogonal

$$\therefore 2gg_1 + 2ff_1 = 0 - 8$$

$$\Rightarrow 2(-k) + (-3) \times 2k = 0 - 8$$

$$\therefore k = 1$$

43. Find the equation of the circle which cuts the circle $x^2 + y^2 - 14x - 8y + 64 = 0$ and the coordinate axes orthogonally.

उस वृत्त का समीकरण ज्ञात करें जो वृत्त $x^2 + y^2 - 14x - 8y + 64 = 0$ तथा निर्देशांक अक्षों को लम्बवत् प्रतिच्छेद करता है।

Ans. $x^2 + y^2 = 64$

Sol. $x^2 + y^2 - 14x - 8y + 64 = 0$

$$(g, f) = (7, 4); c_1 = 64$$

$$\text{Let } x^2 + y^2 = a^2$$

$$\Rightarrow x^2 + y^2 - a^2 = 0$$

$$(g, f) = (0, 0); c_2 = -a^2$$

$$2g_1g_2 + 2f_1f_2 = c_1 + c_2$$

$$0 + 0 = 64 + (-a^2)$$

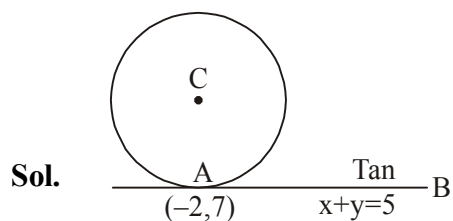
$$a^2 = 64$$

$$\text{i.e., } x^2 + y^2 = 64$$

44. Find the equation of a circle which touches the line $x + y = 5$ at the point $(-2, 7)$ and cuts the circle $x^2 + y^2 + 4x - 6y + 9 = 0$ orthogonally.

रेखा $x + y = 5$ को बिन्दु $(-2, 7)$ पर स्पर्श करने वाले तथा वृत्त $x^2 + y^2 + 4x - 6y + 9 = 0$ को लम्बकोणीय प्रतिच्छेद करने वाले वृत्त का समीकरण ज्ञात कीजिए।

Ans. $x^2 + y^2 + 7x - 11y + 38 = 0$



$$(x + 2)^2 + (y - 7)^2 + \lambda(x + y - 5) = 0$$

$$\text{or } x^2 + y^2 + x(4 + \lambda) + y(-14 + \lambda) + (53 + 5\lambda) = 0$$

It cuts orthogonally the circle

$$2 \frac{4+\lambda}{2} \cdot 2 + 2 \frac{(-14+\lambda)}{2} (-3) = 9 + 53 - 5\lambda$$

$$\text{or } \lambda(2 - 3 + 5) = 62 - 42 - 8$$

$$\text{or } 4\lambda = 12 \Rightarrow \lambda = 3$$

Hence the required circle is

$$x^2 + y^2 + 7x - 11y + 38 = 0$$

45. Find the equation of the circle passing through the point $(-6, 0)$ if the power of the point $(1, 1)$ w.r.t. the circle is 5 and it cuts the circle $x^2 + y^2 - 4x - 6y - 3 = 0$ orthogonally.

बिन्दु $(-6, 0)$ से गुजरने वाले वृत्त का समीकरण ज्ञात करें, यदि बिन्दु $(1, 1)$ का वृत्त के सापेक्ष शक्ति 5 है तथा यह वृत्त, वृत्त $x^2 + y^2 - 4x - 6y - 3 = 0$ के लम्बकोणीय प्रतिच्छेद करता है।

Ans. $x^2 + y^2 + 6x - 3y = 0$

Sol. Let the circle be $x^2 + y^2 + 2gx + 2fy + c = 0$

$(-6, 0)$ lies on the circle.

$$\therefore 36 - 12g + c = 0 \quad \text{_____ (i)}$$

The power of $(1, 1)$ is 5.

$$\Rightarrow 1 + 1 + 2g + 2f + c = 5$$

$$\Rightarrow 2g + 2f + c = 3 \quad \text{_____ (ii)}$$

S is orthogonal to

$$x^2 + y^2 - 4x - 6y - 3 = 0$$

$$\Rightarrow 2g(-2) + 2f(-3) = c - 3$$

$$\therefore 4g + 6f + c - 3 = 0 \quad \text{_____ (iii)}$$

From (ii) and (iii)

$$2g + 2c = 6 \Rightarrow g + c = 3 \quad \text{_____ (iv)}$$

From (i) and (iv) $g = 3$

$$\Rightarrow c = 0 \quad \& \quad f = \frac{-3}{2}$$

$$s \equiv x^2 + y^2 + 6x - 3y = 0$$

46. Find the equation of the circle which passes through the points of intersection of circles $x^2 + y^2 - 2x - 6y + 6 = 0$ and $x^2 + y^2 + 2x - 6y + 6 = 0$ and intersects the circle $x^2 + y^2 + 4x + 6y + 4 = 0$ orthogonally.

वृत्तों $x^2 + y^2 - 2x - 6y + 6 = 0$ व $x^2 + y^2 + 2x - 6y + 6 = 0$ के प्रतिच्छेद बिन्दुओं से गुजरने वाले उस वृत्त समीकरण ज्ञात करें जो वृत्त $x^2 + y^2 + 4x + 6y + 4 = 0$ को लम्बकोणीय प्रतिच्छेद करता है।

Ans. $x^2 + y^2 + 14x - 6y + 6 = 0$

Sol. By $S_1 + \lambda S_2$ the required circle is

$$(x^2 + y^2 - 2x + 6y + 6) + \lambda(x^2 + y^2 + 2x - 6y + 6) = 0$$

$$\text{or } (x^2 + y^2 - 6y + 6)(1 + \lambda) + 2x(\lambda - 1) = 0$$

$$\text{or } x^2 + y^2 + 2x \frac{\lambda - 1}{\lambda + 1} - 6y + 6 = 0 \quad \text{_____}(1)$$

It cuts $x^2 + y^2 + 4x + 6y + 4 = 0$ orthogonally.

$$\text{Hence, } 2g_1g_2 + 2f_1f_2 = c_1 + c_2$$

$$\text{i.e. } 2 \frac{\lambda - 1}{\lambda + 1} \cdot 2 - 2 \cdot 3 \cdot 3 = 6 + 4$$

$$\text{or } 4(\lambda - 1) = 28(\lambda + 1)$$

$$\lambda - 1 = 7\lambda + 7 \quad \therefore \lambda = \frac{-4}{3}$$

Hence the required circle by (1) is

$$x^2 + y^2 + 14x - 6y + 6 = 0$$

47. Line $2x + 3y + 1 = 0$ is a tangent to a circle at $(1, -1)$. This circle is orthogonal to a circle which is drawn having diameter as a line segment with end points $(0, -1)$ and $(-2, 3)$. Find equation of circle.

रेखा $2x + 3y + 1 = 0$ एक वृत्त को $(1, -1)$ पर स्पर्श करती है। यह वृत्त एक ऐसे वृत्त को लम्बकोणीय प्रतिच्छेद करता है जो कि रेखाखण्ड जिसके सिरे $(0, -1)$ व $(-2, 3)$ को व्यास लेकर खींचा गया है। वृत्त का समीकरण ज्ञात कीजिए।

Ans. $2x^2 + 2y^2 - 10x - 5y + 1 = 0$

Sol. The required circle is $(x - 1)^2 + (y + 1)^2 + \lambda(2x + 3y + 1) = 0$

$$\text{or } x^2 + y^2 + 2(\lambda - 1)x + (3\lambda + 2)y + \lambda + 2 = 0 \quad \text{_____}(1)$$

The circle having the line joining $(0, -1)$ and $(-2, 3)$ as diameter is

$$x(x + 2) + (y + 1)(y - 3) = 0$$

$$\text{or } x^2 + y^2 + 2x - 2y - 3 = 0 \quad \text{_____}(2)$$

Equations (1) and (2) are orthogonal.

$$\therefore 2(\lambda - 1) - (3\lambda + 2) = \lambda + 2 - 3$$

On simplifying,

$$\Rightarrow \lambda = -\frac{3}{2}$$

Hence equation (1) becomes

$$x^2 + y^2 + 2\left(-\frac{3}{2} - 1\right)x + \left(3 \times -\frac{3}{2} + 2\right)y - \frac{3}{2} + 2 = 0$$

$$\Rightarrow x^2 + y^2 - 5x - \frac{5}{2}y + \frac{1}{2} = 0$$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question nos. 48 to 50

Two fixed points A and B are 4 units apart, and are on the same side of a moving line L. If perpendicular distances of A and B say p_1 and p_2 from the line L are such that $p_1 + 3p_2 = k$, k being a constant, then the line L always touches a fixed circle C.

दो स्थिर बिन्दु A एवं B के बीच की दूरी 4 है तथा यह एक गतिशील रेखा L के एक ही तरफ है यदि A एवं B की रेखा L से दूरी p_1 एवं p_2 है जहाँ $p_1 + 3p_2 = k$ (स्थिरांक) तो रेखा L हमेशा एक स्थिर वृत्त C को स्पर्श करेगी।

48. The centre of the circle C lies on

(A) line segment joining AB

(B) perpendicular bisector of AB

(C) one of A or B

(D) nothing definite can be said

C का केन्द्र होगा :

(A) ल रेखाखण्ड पर

(B) ल के लम्बाधक पर

(C) या तो A या B पर

(D) कुछ निश्चित नहीं कहा जा सकता

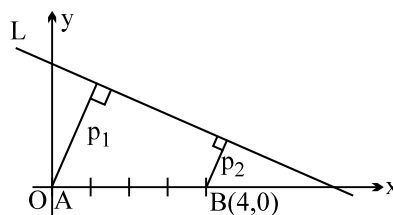
Ans. A

Sol. Let A = (0, 0) and B = (4, 0)
and the line be $ax + by = 1$

$$p_1 = \left| \frac{1}{\sqrt{a^2 + b^2}} \right| ; \quad p_2 = \left| \frac{4a - 1}{\sqrt{a^2 + b^2}} \right|$$

$$p_1 + 3p_2 = k$$

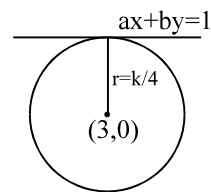
$$\left| \frac{1}{\sqrt{a^2 + b^2}} \right| + 3 \left| \frac{4a - 1}{\sqrt{a^2 + b^2}} \right| = k$$



now (0, 0) and (4, 0) must give the same sign i.e. – ve with the line L ($4a - 1 < 0$)

$$\therefore \frac{1}{\sqrt{a^2 + b^2}} + \frac{3(1-4a)}{\sqrt{a^2 + b^2}} = k$$

$$\left| \frac{4(1-3a)}{\sqrt{a^2 + b^2}} \right| = k ; \quad \left| \frac{3a-1}{\sqrt{a^2 + b^2}} \right| = \frac{k}{4}$$



hence centre of the fixed circle is (3, 0) which lies on the line segment AB $\Rightarrow$ (A)

49. If $k = 4$ then the radius of the circle is :

यदि $k = 4$ तो वृत्त C की त्रिज्या होगी :

- (A) 1 (B) 2 (C) 4 (D) 8

Ans. A

Sol. If $k = 4 \Rightarrow r = 1$ Ans.

50. If A and B are $(-2, 0)$ and $(2, 0)$ respectively, then the centre of the circle C is :

यदि A $(-2, 0)$ तथा B $(2, 0)$ तो C की त्रिज्या होगी :

- (A) (0, 1) (B) (1, 0) (C) $(3/2, 0)$ (D) can not be found

Ans. B

Sol. $p_1 = \left| \frac{-2a-1}{\sqrt{a^2 + b^2}} \right| ; \quad p_2 = \left| \frac{2a-1}{\sqrt{a^2 + b^2}} \right|$

hence, with the same argument, $2a - 1 < 0$

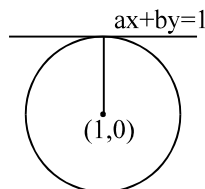
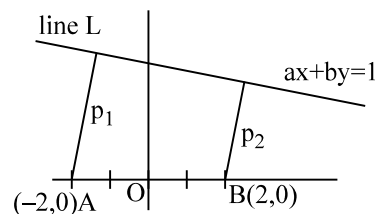
and $-2a - 1 < 0$

$$\therefore p_1 + 3p_2 = k$$

$$\frac{1+2a}{\sqrt{a^2 + b^2}} + \frac{3(1-2a)}{\sqrt{a^2 + b^2}} = k$$

$$\frac{4(1-a)}{\sqrt{a^2 + b^2}} = k$$

$$\frac{|a-1|}{\sqrt{a^2 + b^2}} = \frac{k}{4}$$



hence centre is (1, 0) Ans.

Paragraph for Question Nos. 51 to 53

A circle C of radius 1 is inscribed in an equilateral triangle PQR. The points of contact of C with the sides PQ, QR, RP are D, E, F respectively. The line PQ is given by the equation $\sqrt{3}x + y - 6 = 0$ and the point D is $\left(\frac{3\sqrt{3}}{2}, \frac{3}{2}\right)$.

Further, it is given that the origin and the centre of C are on the same side of the line PQ.

समबाहु त्रिभुज PQR के अन्दर एक वृत्त स्थित है जिसकी त्रिज्या 1 है। वृत्त C के त्रिभुज की भुजाओं PQ, QR, RP के साथ स्पर्श

बिन्दु क्रमशः D, E, F हैं। रेखा PQ का समीकरण $\sqrt{3}x + y - 6 = 0$ तथा बिन्दु D $\left(\frac{3\sqrt{3}}{2}, \frac{3}{2}\right)$ है। यह भी दिया है कि मूलबिन्दु व

C का केन्द्र रेखा PQ के एक तरफ ही स्थित है।

51. The equation of circle C is :

[JEE-2008]

वृत्त C का समीकरण है :

[JEE-2008]

(A) $(x - 2\sqrt{3})^2 + (y - 1)^2 = 1$

(B) $(x - 2\sqrt{3})^2 + (y + \frac{1}{2})^2 = 1$

(C) $(x - \sqrt{3})^2 + (y + 1)^2 = 1$

(D) $(x - \sqrt{3})^2 + (y - 1)^2 = 1$

Ans. D

Sol. $m_{PQ} = -\sqrt{3}$

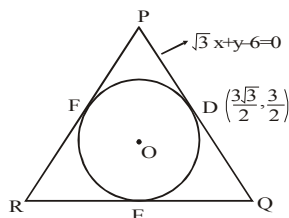
so slope of OD = $\frac{1}{\sqrt{3}}$

$\tan \theta = \frac{1}{\sqrt{3}}$

$$\therefore \frac{x - \frac{3\sqrt{3}}{2}}{\frac{\sqrt{3}}{2}} = \frac{y - \frac{3}{2}}{\frac{1}{2}} = \pm 1$$

$(2\sqrt{3}, 2)$ (not possible) and $(\sqrt{3}, 1)$

hence circle is $(x - \sqrt{3})^2 + (y - 1)^2 = 1$



52. Points E and F are given by :

[JEE-2008]

बिन्दु E व F होंगे :

[JEE-2008]

$$(A) \left(\frac{\sqrt{3}}{2}, \frac{3}{2} \right), (\sqrt{3}, 0)$$

$$(B) \left(\frac{\sqrt{3}}{2}, \frac{1}{2} \right), (\sqrt{3}, 0)$$

$$(C) \left(\frac{\sqrt{3}}{2}, \frac{3}{2} \right), \left(\frac{\sqrt{3}}{2}, \frac{1}{2} \right)$$

$$(D) \left(\frac{3}{2}, \frac{\sqrt{3}}{2} \right), \left(\frac{\sqrt{3}}{2}, \frac{1}{2} \right)$$

Ans. A

Sol. For point E $\frac{x-\sqrt{3}}{-\frac{\sqrt{3}}{2}} = \frac{y-1}{\frac{1}{2}} = 1 \left[\therefore E \left(\frac{\sqrt{3}}{2}, \frac{3}{2} \right) \right]$

For point F $\frac{x-\sqrt{3}}{0} = \frac{y-1}{-1} = 1 \left[\therefore F(\sqrt{3}, 0) \right]$

53. Equations of the sides RP, RQ are :

[JEE-2008]

भुजाओं RP व RQ के समीकरण होंगे :

[JEE-2008]

$$(A) y = \frac{2}{\sqrt{3}}x + 1, y = -\frac{2}{\sqrt{3}}x - 1$$

$$(B) y = \frac{1}{\sqrt{3}}x, y = 0$$

$$(C) y = \frac{\sqrt{3}}{2}x + 1, y = -\frac{\sqrt{3}}{2}x - 1$$

$$(D) y = \sqrt{3}x, y = 0$$

Ans. D

Sol. Equation of line RP $y = 0$

Equation of line QR $y - \frac{3}{2} = \sqrt{3} \left(x - \frac{\sqrt{3}}{2} \right)$

$$y = \sqrt{3}x$$

Paragraph for Question Nos. 54 to 56

Let S_1 , S_2 and S_3 be the circles $x^2 + y^2 = 4$, $x^2 + y^2 - 6x + 5 = 0$ and $x^2 + y^2 - 4y - 5 = 0$ respectively, then

माना कि वृत्त S_1 , S_2 तथा S_3 क्रमशः $x^2 + y^2 = 4$, $x^2 + y^2 - 6x + 5 = 0$ एवं $x^2 + y^2 - 4y - 5 = 0$ है, तो

54. Point (2, 2) lies :

(A) in the circle S_1 only

(B) in the circle S_2 only

(C) in the circle S_3 only

(D) in the circles S_1 and S_2 but not in S_3

बिन्दु (2, 2) स्थित है :

(A) केवल वृत्त S_1 के अन्दर

(B) केवल वृत्त S_2 के अन्दर

(C) केवल वृत्त S_3 के अन्दर

(D) वृत्त S_1 तथा S_2 के अन्दर लेकिन S_3 के अन्दर नहीं

Ans. C

Sol. $S_3 < 0$

55. The point (1, 2) lies :

(A) in circles S_2 and S_3 but not in circle S_1

(B) in circles S_2 and S_1 but not in circle S_3

(C) in circle S_3 but not in circles S_1 and S_2

(D) in circles S_1, S_2 and S_3

बिन्दु (1, 2) स्थित है :

(A) वृत्त S_2 तथा S_3 के अन्दर लेकिन S_1 के अन्दर नहीं

(B) वृत्त S_2 तथा S_1 के अन्दर लेकिन S_3 के अन्दर नहीं

(C) वृत्त S_3 के अन्दर लेकिन S_2 तथा S_1 के अन्दर नहीं

(D) वृत्त S_1, S_2 तथा S_3 के अन्दर

Ans. C

Sol. $S_3 < 0$

56. The line $x - y + 2\sqrt{2} = 0$ is :

(A) tangent to S_1 and does not meet S_2 and S_3

(B) tangent to S_1 , secant of S_3 and does not meet S_2

(C) tangent to S_1 and secant of S_2 and S_3

(D) tangent to S_3 and secant of S_1 and S_2

रेखा $x - y + 2\sqrt{2} = 0$:

(A) S_1 की स्पर्श रेखा है तथा S_2 एवं S_3 को नहीं मिलती है

(B) S_1 की स्पर्श रेखा है, S_3 की जीवा है तथा S_2 को नहीं मिलती है

(C) S_1 की स्पर्श रेखा है तथा S_2 एवं S_3 की जीवा है

(D) S_3 की स्पर्श रेखा है तथा S_1 एवं S_2 की जीवा है

Ans. B

Sol. For $S_1 \rightarrow P = r$, $S_2 \rightarrow P > r$, $S_3 \rightarrow P < r$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

57. A circle passes through the points $(-1, 1)$, $(0, 6)$ and $(5, 5)$. The point(s) on this circle, the tangent(s) at which is/are parallel to the straight line joining the origin to its centre is/are :

एक वृत्त बिन्दुओं $(-1, 1)$, $(0, 6)$ तथा $(5, 5)$ से गुजरता है। वृत्त पर स्थित बिन्दु जहाँ पर स्पर्श रेखा मूल बिन्दु को केन्द्र से जोड़ने वाली

- (A) $(1, -5)$ (B) $(5, 1)$ (C) $(-5, -1)$ (D) $(-1, 5)$

Ans. BD

Sol. Note that Δ is right angled at $(0, 6)$. Centre of the circle is $(2, 3)$. Slope of the line joining origin to the centre is $3/2$. Take parametric equation of a line through $(2, 3)$ with

$$\tan \theta = -\frac{2}{3} \text{ as } \frac{x-2}{\cos \theta} = \frac{y-3}{\sin \theta} = \pm r \text{ where } r = \sqrt{13}.$$

Get the co-ordinates on the circle

58. Three distinct lines are drawn in a plane. Suppose there exist exactly n circles in the plane tangent to all the three lines, then the possible values of n is/are :

किसी तल में तीन भिन्न-भिन्न रेखाएँ है माना कि उस तल में कुल n वृत्त है जो कि तीनों रेखाओं को स्पर्श करती है तो n का सम्भावित

- (A) 0 (B) 1 (C) 2 (D) 4

Ans. ACD

Sol. Case-1: If lines form a triangle then $n = 4$

i.e. 3 excircles and 1 incircle

Case-2: If lines are concurrent

or all 3 parallel then $n = 0$

Case-3: If two are parallel

and third cuts then $n = 2$

hence (A), (C), (D)

59. Equations of circles which pass through the points $(1, -2)$ and $(3, -4)$ and touch the x -axis is :

- (A) $x^2 + y^2 + 6x + 2y + 9 = 0$ (B) $x^2 + y^2 + 10x + 20y + 25 = 0$
(C) $x^2 + y^2 - 6x + 4y + 9 = 0$ (D) none

बिन्दुओं $(1, -2)$ एवं $(3, -4)$ से गुजरने वाले एवं x -अक्ष को स्पर्श करने वाले वृत्त का समीकरण है :

- (A) $x^2 + y^2 + 6x + 2y + 9 = 0$ (B) $x^2 + y^2 + 10x + 20y + 25 = 0$
(C) $x^2 + y^2 - 6x + 4y + 9 = 0$ (D) इनमें से कोई नहीं

Ans. BC

Sol. Let equation of required circle is

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

it passes through $(1, -2)$ & $(3, -4)$

$$2g - 4f + c = -5$$

$$6g - 8f + c = -25$$

$$4g - 8f + 2c = -10$$

$$6g - 8f + c = -25$$

$$-2g + c = 15$$

$$\chi\rho\chi\lambda\epsilon\tau\omicron\upsilon\chi\eta\epsilon\sigma\xi-\alpha\xi\iota\sigma\gamma^2 = c \Rightarrow$$

$$g^2 - 2g - 15 = 0$$

$$g = 5, -3$$

$$g = 5, c = 25, f = 10 \Rightarrow x^2 + y^2 + 10x + 20y + 25 = 0$$

$$g = -3, c = 9, f = 2 \Rightarrow x^2 + y^2 - 6x + 4y + 9 = 0$$

60. The circle $x^2 + y^2 - 2x - 3ky - 2 = 0$ passes through two fixed points :

वृत्त $x^2 + y^2 - 2x - 3ky - 2 = 0$ दो स्थिर बिन्दुओं से गुजरता है, वे हैं :

- (A) $(1 + \sqrt{3}, 0)$ (B) $(-1 + \sqrt{3}, 0)$ (C) $(-\sqrt{3} - 1, 0)$ (D) $(1 - \sqrt{3}, 0)$

Ans. AD

Sol. Two fixed pts. are point of intersection of

$$x^2 + y^2 - 2x - 2 = 0 \quad \& \quad y = 0$$

$$\text{Point } x^2 - 2x - 2 = 0$$

$$(x - 1)^2 - 3 = 0 \Rightarrow x - 1 = \sqrt{3}, x - 1 = -\sqrt{3} \Rightarrow (1 + \sqrt{3}, 0)(1 - \sqrt{3}, 0)$$

61. The equation of the circle which touches both the axes and the line $\frac{x}{3} + \frac{y}{4} = 1$ and lies in the first quadrant is

$$(x - c)^2 + (y - c)^2 = c^2 \text{ where } c \text{ is :}$$

प्रथम चतुर्थांश में दोनों अक्षों तथा सरल रेखा $\frac{x}{3} + \frac{y}{4} = 1$ को स्पर्श करने वाले वृत्त का समीकरण $(x - c)^2 + (y - c)^2 = c^2$ है जहाँ c

का मान है :

- (A) 1 (B) 2 (C) 4 (D) 6

Ans. AD

$$\text{Sol. } \left| \frac{4C + 3C - 12}{5} \right| = C \Rightarrow C = 1, 6$$

62. Tangents PA and PB are drawn to the circle $S \equiv x^2 + y^2 - 2y - 3 = 0$ from the point P(3, 4). Which of the following alternative(s) is/are correct?

- (A) The power of point P(3, 4) with respect to circle $S = 0$ is 14

(B) The angle between tangents from $P(3, 4)$ to the circle $S = 0$ is $\frac{\pi}{3}$

(C) The equation of circumcircle of ΔPAB is $x^2 + y^2 - 3x - 5y + 4 = 0$

(D) The area of quadrilateral $PACB$ is $3\sqrt{7}$ square units where C is the centre of circle $S = 0$

वृत्त $S \equiv x^2 + y^2 - 2y - 3 = 0$ पर बिन्दु $P(3, 4)$ से खींची गयी स्पर्श रेखा PA एवं PB है तो सही विकल्प चुनियें?

(A) $\sin \theta > 0$ का बिन्दु $P(3, 4)$ के सापेक्ष बिन्दु क्षमता 14 है

(B) $\angle APB$ एवं PB के बीच कोण $\frac{\pi}{3}$ है

(C) ΔPAB का परिवृत्त $x^2 + y^2 - 3x - 5y + 4 = 0$ है

(D) चतुर्भुज $PACB$ का क्षेत्रफल $3\sqrt{7}$ है जहाँ C वृत्त का केन्द्र है

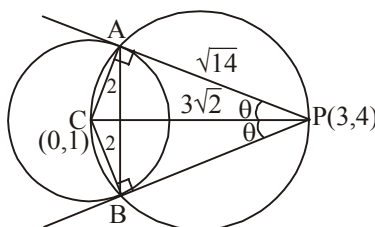
Ans. AC

Sol. Clearly power of point $P(3, 4)$ w.r.t. $S = 0$ is

$$S_1 = 14 \Rightarrow \text{option (A) is correct}$$

$$\text{Also } \sin \theta = \frac{2}{3\sqrt{2}} = \frac{\sqrt{2}}{3}$$

$$\text{Now angle between tangents} = 2 \sin^{-1} \frac{\sqrt{2}}{3}$$



Also CP is diameter of circumcircle of ΔPAB , so equation of circumcircle is

$$x(x-3) + (y-1)(y-4) = 0 \Rightarrow x^2 + y^2 - 3x - 5y + 4 = 0 \Rightarrow \text{option (C) is correct}$$

Now, area of quadrilateral $PACB = 2 \times \frac{1}{2} \times 2 \times \sqrt{14} = 2\sqrt{14}$ square units. $\Rightarrow$ option (D) is incorrect

63. If the equation of circle touching the y -axis at $(0, 3)$ and making an intercept of 8 unit on x -axis is $x^2 + y^2 + 2gx + 2fy + c = 0$, then $(g + f + c)$ can be :

यदि वृत्त $x^2 + y^2 + 2gx + 2fy + c = 0$ y -अक्ष को $(0, 3)$ पर स्पर्श कर रही है तथा x -अक्ष पर 8 लम्बाई का अन्तःखण्ड काट रही है r $(g + f + c)$ होगा :

(A) 1 (B) 7 (C) 11 (D) 14

Ans. AC

Sol. The circle is $x^2 + (y-3)^2 + \lambda x = 0$

putting $y = 0$, we get $x^2 + \lambda x + 9 = 0$

x -intercept is $|x_1 - x_2|$

$$|x_1 - x_2| = \sqrt{(x_1 + x_2)^2 - 4x_1x_2}$$

$$64 = \lambda^2 - 36 \Rightarrow \lambda^2 = 100 \Rightarrow \lambda = 10 \text{ or } -10$$

Solving $\lambda = \pm 10$

The circles are $x^2 + y^2 \pm 10x - 6y + 9 = 0$

$\Rightarrow g = \pm 5; f = -3; c = 9 \Rightarrow (g + f + c) = 11$ and 1 Ans.

64. The circles $x^2 + y^2 - 2x - 4y + 1 = 0$ and $x^2 + y^2 + 4x + 4y - 1 = 0$
- (A) touch internally
(B) touch externally
(C) have $3x + 4y - 1 = 0$ as the common tangent at the point of contact
(D) have $3x + 4y + 1 = 0$ as the common tangent at the point of contact

वृत्तों $x^2 + y^2 - 2x - 4y + 1 = 0$ तथा $x^2 + y^2 + 4x + 4y - 1 = 0$ के लिये सत्य होगा :

- (A) आन्तरिक स्पर्श करते हैं
(B) बाह्य स्पर्श करते हैं
(C) इनकी उभयनिष्ठ स्पर्शरेखा $3x + 4y - 1 = 0$ है
(D) इनकी उभयनिष्ठ स्पर्शरेखा $3x + 4y + 1 = 0$ है

Ans. BC

Sol. Correct options are (B) and (C).

For given circles $S_1 : x^2 + y^2 - 2x - 4y + 1 = 0$

and $S_2 : x^2 + y^2 + 4x + 4y - 1 = 0$

$C_1 (1, 2), r_1 = 2$ and $C_2 (-2, -2), r_2 = 3$

Now, $r_1 + r_2 = 5$ and $C_1 C_2 = 5$

Hence, circles touch externally.

Also common tangent at point of contact is

$S_1 - S_2 = 0$ or $3x + 4y - 1 = 0$

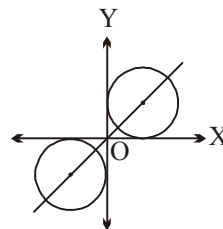
65. Which of the following is/are True? The circles $x^2 + y^2 - 6x - 6y + 9 = 0$ and $x^2 + y^2 + 6x + 6y + 9 = 0$ are such that:
- (A) they do not intersect (B) they touch each other.
(C) their exterior common tangents are parallel (D) their interior common tangents are perpendicular

वृत्त $x^2 + y^2 - 6x - 6y + 9 = 0$ तथा $x^2 + y^2 + 6x + 6y + 9 = 0$ के लिये कौनसा विकल्प सत्य है :

- (A) परस्पर काटते नहीं (B) परस्पर स्पर्श करते हैं
(C) बाह्य स्पर्श रेखायें परस्पर समान्तर हैं (D) आन्तरिक स्पर्श रेखायें परस्पर लम्बवत् हैं

Ans. ACD

Sol. Centre (3, 3) with radius = 3 and centre (−3, 3) with radius = 3.



66. Consider the circles $C_1: x^2 + y^2 = 16$ and $C_2: x^2 + y^2 - 12x + 32 = 0$. Which of the following statements is/are correct?

- (A) Number of common tangent to these circles is 3.
- (B) The point P with coordinates (4, 1) lies outside the circle C_1 and inside the circle C_2 .
- (C) Their direct common tangent intersect at (12, 0).
- (D) Slope of their radical axis is not defined.

यदि $C_1: x^2 + y^2 = 16$ तथा $C_2: x^2 + y^2 - 12x + 32 = 0$ तो इनमें से कौनसा विकल्प सत्य है?

- (A) कुल उभयनिष्ठ स्पर्शरेखा की संख्या 3 होगी
- (B) च ; 4ए1द्वृत C_1 के बाहर तथा C_2 के अन्दर होगा
- (C) उनकी उभयनिष्ठ बाह्य स्पर्श रेखा (12, 0) पर काटेगी
- (D) इनके मूलाक्ष की प्रवणता अपरिभाषित होगी

Ans. ACD

Sol. $C_1: (0, 0)$, radius 4
 $C_2: (6, 0)$, radius 2
 circle touch externally.

67. If a circle passes through P(0, 1), Q(0, 9) and touches the x – axis, then which of the following statement(s) is/are TRUE?

- (A) Centres of circles are $(\pm 3, \pm 5)$.
- (B) Equation of one of their direct common tangent is $y = 10$.
- (C) Radii of both the circles is 5.
- (D) Length of common chord of circles is 8.

बिन्दुओं P(0, 1), Q(0, 9) से जाने वाला तथा x – अक्ष को स्पर्श करने वाले वृत्त के लिये निम्न से कौनसा कथन सत्य है?

- (A) केन्द्र $(\pm 3, \pm 5)$ होगा
- (B) उनकी बाह्य उभयनिष्ठ स्पर्शरेखा में से एक रेखा $y = 10$ होगी
- (C) दोनों वृत्त की त्रिज्या 5 हो
- (D) उभयनिष्ठ जीवा की लम्बाई 8 हो

Ans. BCD

Sol. Equation of any circle through P(0, 1) and Q(0, 9) is $x^2 + (y - 1)(y - 9) + \lambda x = 0$

$$\Rightarrow x^2 + y^2 - 10y + 9 + \lambda x = 0$$

As above circle touches x - axis, so put $y = 0$

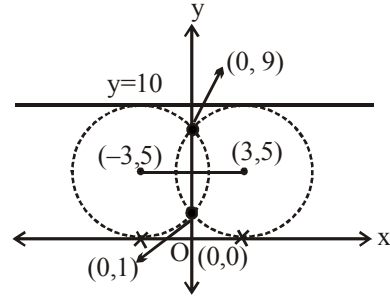
$$\Rightarrow x^2 + \lambda x + 9 = 0$$

should have equal roots. So, Discriminant = 0

$$\Rightarrow \lambda^2 = 36 \quad \therefore \lambda = \pm 6$$

Hence required circles are $x^2 + y^2 + 6x - 10y + 9 = 0$

and $x^2 + y^2 - 6x - 10y + 9 = 0$



68. A family of linear functions is given by $f(x) = 1 + c(x + 3)$ where $c \in \mathbb{R}$. If a member of this family meets a unit circle centred at origin in two coincident points then 'c' can be equal to :

एक रेखीय फलनों का समूह $f(x) = 1 + c(x + 3)$ द्वारा दिया जाता है। यदि समूह का एक सदस्य इकाई त्रिज्या तथा मूल बिन्दु केन्द्रित एक वृत्त के दो सङ्गति बिन्दुओं पर मिलेगा :

- (A) $-3/4$ (B) 0 (C) $3/4$ (D) 1

Ans. AB

Sol. Line: $y = 1 + c(x + 3)$ is tangent to circle $x^2 + y^2 = 1$

$$\therefore p = r$$

$$\therefore \left| \frac{1+3c}{\sqrt{1+c^2}} \right| = 1$$

$$(1 + 3c)^2 = 1 + c^2$$

$$9c^2 + 6c = c^2$$

$$8c^2 = -6c$$

$$\therefore c = 0 \quad \text{or} \quad c = -3/4 \Rightarrow (A), (B)$$

69. Which of the following lines have the intercepts of equal lengths on the circle, $x^2 + y^2 - 2x + 4y = 0$?

निम्न में से कौनसी रेखाएँ वृत्त $x^2 + y^2 - 2x + 4y = 0$ पर बराबर लम्बाई के अन्तः कर्ण काटती हैं?

- (A) $3x - y = 0$ (B) $x + 3y = 0$ (C) $x + 3y + 10 = 0$ (D) $3x - y - 10 = 0$

Ans. ABCD

Sol. Chords equidistance from the centre are equal

70. The equations of the tangents drawn from the origin to the circle, $x^2 + y^2 - 2rx - 2hy + h^2 = 0$ are :

वृत्त $x^2 + y^2 - 2rx - 2hy + h^2 = 0$ पर मूल बिन्दु से डाली गई स्पर्श रेखाएँ हैं :

- (A) $x = 0$ (B) $y = 0$
(C) $(h^2 - r^2)x - 2rhy = 0$ (D) $(h^2 - r^2)x + 2rhy = 0$

Ans. AC

Sol. $x^2 + y^2 - 2rx - 2hy + h^2 = 0$

$$x^2 + y^2 - 2rx - 2hy + h^2 + r^2 = r^2$$

$$(x - r)^2 + (y - h)^2 = r^2$$

(r, h) is centre, r is radius

It is touching $x = 0$ because radius Δx .

coordinate of centre are equal

$x = 0$ is a tangent

Let $y = mx$ be another tangent [$y - mx = 0$]

1 distance from (r,h) = r.

$$\left| \frac{r - mh}{\sqrt{1 + m^2}} \right| = r \quad m = 0, \quad \frac{h^2 - r^2}{2rh}$$

Another tangent will be $(h^2 - r^2)x - 2rhy = 0$

B is correct.

MATCH THE COLUMN TYPE QUESTION(S)

71. Match the following column :

Column – I

Column – II

- | | |
|---|-------|
| (A) Number of values of a for which the common chord of the circles $x^2 + y^2 = 8$ and $(x - a)^2 + y^2 = 8$ subtends a right angle at the origin is | (P) 4 |
| (B) A chord of the circle $(x - 1)^2 + y^2 = 4$ lies along the line $y = 22\sqrt{3}(x - 1)$. The length of the chord is equal to | (Q) 2 |
| (C) The number of circles touching all the three lines $3x + 7y = 2$, $21x + 49y = 5$ and $9x + 21y = 0$ are | (R) 0 |
| (D) If radii of the smallest and largest circle passing through the point $(\sqrt{3}, \sqrt{2})$ and touching the circle $x^2 + y^2 - 2\sqrt{2}y - 2 = 0$ are r_1 and r_2 respectively, then the mean of r_1 and r_2 is | (S) 1 |

निम्न कॉलम को सुमेलित कीजिए :

Column – I

Column – II

- | | |
|--|-------|
| (A) a के उन मानों की संख्या जिनके लिए वृत्तों $x^2 + y^2 = 8$ तथा $(x - a)^2 + y^2 = 8$ की उभयनिष्ठ जीवा मूल बिन्दु पर समकोण अन्तरित करती है | (P) 4 |
| (B) यदि वृत्त $(x - 1)^2 + y^2 = 4$ की एक जीवा रेखा $y = 22\sqrt{3}(x - 1)$ के अनुदिश हो, तो जीवा की लम्बाई है | (Q) 2 |

(C) सभी तीन रेखाओं $3x + 7y = 2$, $21x + 49y = 5$ तथा $9x + 21y = 0$ को (R) 0
स्पर्श करने वाले वृत्तों की संख्या है

(D) यदि बिन्दु $(\sqrt{3}, \sqrt{2})$ से गुजरने वाले तथा वृत्त $x^2 + y^2 - 2\sqrt{2}y - 2 = 0$ को (S) 1

स्पर्श करने वाले सबसे छोटे तथा सबसे बड़े वृत्त की त्रिज्याएँ क्रमशः r_1 तथा r_2
हो, तो r_1 तथा r_2 का माध्य है

Ans. (A) $\rightarrow Q$; (B) $\rightarrow P$; (C) $\rightarrow R$; (D) $\rightarrow S$

Sol. (A) $S_1 - S_2 = 0$ is the required common chord i.e. $2x = a$

Make homogeneous, we get $x^2 + y^2 - 8.4 \frac{x^2}{a^2} = 0$

As pair of lines subtending angle of 90° at origin

$\therefore$ coefficient of $x^2 +$ coefficient of $y^2 = 0$

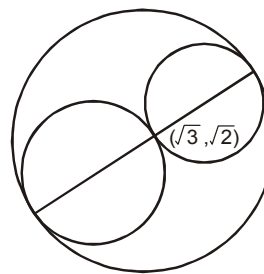
$\therefore a = \pm 4$

(B) $y = 22\sqrt{3}(x - 1)$ passes through centre $(1, 0)$ of circle

(C) Three lines are parallel so not any circle is possible

(D) $2(r_1 + r_2) = 4$

$$r_1 + r_2 = 2 \Rightarrow \frac{r_1 + r_2}{2} = 1$$



72. Match the following column :

Column – I

(A) Number of common tangents of the circles

$$x^2 + y^2 - 2x = 0 \text{ and } x^2 + y^2 + 6x - 6y + 2 = 0 \text{ is}$$

(B) Number of transverse common tangents of the circles

$$x^2 + y^2 - 4x - 10y + 4 = 0 \text{ \& } x^2 + y^2 - 6x - 12y - 55 = 0 \text{ is}$$

(C) Number of common tangents of the circles $x^2 + y^2 - 2x - 4y = 0$

$$\text{ \& } x^2 + y^2 - 8y - 4 = 0 \text{ is}$$

(D) Number of direct common tangents of the circles

$$x^2 + y^2 + 2x - 8y + 13 = 0 \text{ \& } x^2 + y^2 - 6x - 2y + 6 = 0 \text{ is}$$

Column – II

(P) 1

(Q) 2

(R) 3

(S) 0

निम्न कॉलम को सुमेलित कीजिए :

Column – I	Column – II
(A) वृत्तो $x^2 + y^2 - 2x = 0$ तथा $x^2 + y^2 + 6x - 6y + 2 = 0$ की उभयनिष्ठ स्पर्श रेखाओं की संख्या है	(P) 1
(B) $x^2 + y^2 - 4x - 10y + 4 = 0$ तथा $x^2 + y^2 - 6x - 12y - 55 = 0$ तिर्यक उभयनिष्ठ स्पर्श रेखाओं की संख्या है	(Q) 2
(C) $x^2 + y^2 - 2x - 4y = 0$ तथा $x^2 + y^2 - 8y - 4 = 0$ की उभयनिष्ठ स्पर्श रेखाओं की संख्या है	(R) 3
(D) $x^2 + y^2 + 2x - 8y + 13 = 0$ तथा $x^2 + y^2 - 6x - 2y + 6 = 0$ की अनुक्रम उभयनिष्ठ स्पर्श रेखाओं की संख्या है	(S) 0

Ans. (A) $\rightarrow$ R; (B) $\rightarrow$ S; (C) $\rightarrow$ P; (D) $\rightarrow$ Q

Sol. (A) $C_1(1, 0)$, $r_1 = 1$ and $C_2(-3, 3)$, $r_2 = 4$
distance between centres C_1 and $C_2 = d = 5$
 $d = r_1 + r_2 = 5 \Rightarrow 3$ common tangents

(B) $C_1(2, 5)$, $r_1 = 5$ and $C_2(3, 6)$, $r_2 = 10$
distance between centres C_1 and $C_2 =$
 $d = \sqrt{2}$
 $d < |r_1 - r_2|$
 $\Rightarrow$ no common tangent

(C) $C_1(1, 2)$, $r_1 = \sqrt{5}$ and $C_2(0, 4)$, $r_2 = 2\sqrt{5}$
distance between centres C_1 and $C_2 = d = \sqrt{5}$
 $|r_1 - r_2| = d$
number of common tangents is 1

(D) $C_1(-1, 4)$, $r_1 = 2$ and $C_2(3, 1)$, $r_2 = 2$
distance between centres C_1 and $C_2 = d = 5$
 $d > r_1 + r_2$
 $\Rightarrow$ number of direct common tangents is 2